TAXATION AND WELFARE IN GHANA: AN AUTOREGRESSIVE DISTRIBUTED LAG (ARDL) APPROACH

A Thesis Submitted in Partial Fulfilment of the Requirements for the Degree of

Master of Philosophy in Accounting

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Declaration

I, XXXXXXX, affirm that this thesis is the result of my own work and has not been presented to any other institution.

Signature:

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Approval

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Abstract

Background: Taxation is a critical instrument for government financing and socio-economic development, funding essential public services including education, healthcare, and social protection programs. However, concerns persist regarding the extent to which taxes influence the welfare of Ghanaians, particularly among vulnerable populations. This study examined the impact of taxation on welfare in Ghana.

Method: The study adopted a quantitative research design using annual time series data from 1985 to 2022 (38 observations) sourced from the Ghana Statistical Service, World Bank World Development Indicators, and the Ghana Revenue Authority. The stationarity properties of the variables were examined using the Augmented Dickey-Fuller (ADF) unit root test, and the Autoregressive Distributed Lag (ARDL) bounds testing approach to cointegration was employed to estimate both short-run and long-run relationships between tax variables and welfare indicators.

Results: Taxes showed no significant effect on income inequality, indicating Ghana's tax system functions primarily for revenue generation rather than redistribution. For education, direct taxes exhibited a significant delayed positive effect, with enrollment improving two years after tax collection. Healthcare access demonstrated strong positive long-run effects, with sustained tax-financed investments expanding service coverage. For poverty, direct taxes had no significant effect, but government expenditure demonstrated immediate poverty-reducing impacts.

Conclusion: Taxation's welfare effects in Ghana are dimension-specific and mediated by government expenditure. While taxation contributes to long-run improvements in education and healthcare access, it does not significantly reduce income inequality. Poverty reduction depends critically on expenditure utilization rather than tax collection alone.

Keywords: Taxation, welfare, income inequality, education, healthcare, poverty reduction.

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List of Abbreviations

- **ADF** - Augmented Dickey-Fuller
- **AIC** - Akaike Information Criterion
- **AG** - Agriculture Value Added
- **ARDL** - Autoregressive Distributed Lag
- **CUSUM** - Cumulative Sum
- **DIT** - Direct Taxes
- **ECM** - Error Correction Model
- **ECT** - Error Correction Term
- **FCUBE** - Free Compulsory Universal Basic Education
- **FDI** - Foreign Direct Investment
- **GDP** - Gross Domestic Product
- **GDPPC** - Gross Domestic Product Per Capita
- **GEX** - Government Expenditure
- **GI** - Gini Index
- **GINI** - Gini Coefficient
- **GRA** - Ghana Revenue Authority
- **GSFP** - Ghana School Feeding Program
- **GSS** - Ghana Statistical Service
- **HCA** - Healthcare Access
- **IDT** - Indirect Taxes
- **INF** - Inflation Rate
- **JB** - Jarque-Bera
- **LEAP** - Livelihood Empowerment Against Poverty
- **LM** - Lagrange Multiplier
- **NARDL** - Nonlinear Autoregressive Distributed Lag
- **NHIS** - National Health Insurance Scheme
- **NONTREV** - Non-Tax Revenue
- **OLS** - Ordinary Least Squares
- **OPN** - Economic Openness
- **PCGDPR** - Per Capita GDP Growth Rate

- **POVH** - Poverty Headcount
- **SD** - Standard Deviation
- **SHS** - Senior High School
- **SSER** - Secondary School Enrollment Rate
- **TDT** - Total Direct Tax Revenue
- **TIT** - Total Indirect Tax Revenue
- **TOT** - Total Tax Revenue
- **TREV** - Total Tax Revenue
- **VAT** - Value-Added Tax
- **VECM** - Vector Error Correction Model
- **WDI** - World Development Indicators

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Dedication

I dedicate this work to God and my family.

Chapter 1 Introduction

1.1 Overview

This chapter provides the foundational background for the study, establishing the rationale and context from which the research problem emerges. It presents the statement of the problem, articulates the aims of the study, and justifies its necessity. Specific research objectives are outlined to guide the investigation, alongside research questions designed to address the identified problem. The chapter further examines the significance of the study to the Ghanaian context, delineates its scope and delimitations, and explains the organization of the thesis. The chapter concludes with a summary that encapsulates its key points and sets the stage for subsequent chapters.

1.2 Background to the Study

Ghana's economic landscape is characterized by traditional agriculture, emerging industries, and a growing services sector (Abbam, Johnson, Dash, & Padmadas, 2018). As one of the most stable and diversified economies in West Africa, Ghana has experienced steady economic growth in recent years (Jaurena, 2018). The country is endowed with rich natural resources such as gold, cocoa, and oil, which contribute significantly to export revenues (Niikoi Kotey et al., 2022). Agriculture remains a major source of employment for the population, while mining, construction, and other industries continue to expand (Eshun, 2018). The services sector, including finance, telecommunications, and tourism, has also experienced notable growth. Despite these

strengths, challenges such as income inequality, infrastructural deficits, and volatility in global commodity prices persist, highlighting the need to strengthen Ghana's economic resilience, sustainability, and inclusiveness.

Taxation serves as a critical instrument for government financing and socio-economic development (Kooshkebaghi, Emamgholipour, & Dargahi, 2022a). It is the primary source of revenue to maintain essential public services, including education, healthcare, social protection, defense, transportation, and public utilities. Beyond resource mobilization, taxation acts as a mechanism for income redistribution, helping to reduce disparities between high- and low-income groups. Effective tax collection enables the government to invest in human capital, improve overall economic conditions, and respond to unforeseen economic shocks, thereby underpinning the nation's socio-economic structure (Kooshkebaghi et al., 2022a; Franklin, Cintya, & Pablo, 2022).

Social welfare plays a pivotal role in promoting inclusive development and ensuring the well-being of citizens (Wokoma, 2023; Jensen & Thompson, 2020). Welfare programs—including social support, healthcare services, education, and other targeted interventions—enhance living standards, foster social cohesion, and reduce inequality (Gal & Bleikh, 2019). Accessible education and healthcare improve human capital, while welfare systems provide protection during periods of economic hardship, ultimately promoting a healthier, better-educated, and more productive population (Sahoo & Choudhury, 2023). Tax systems comprise multiple categories of levies, including personal income tax, value-added tax (VAT), and corporate tax (Andreyeva, Marple, Marinello, Moore, & Powell, 2022; Ayangbah & Sun, 2017). Income tax in Ghana is progressive, with rates ranging from 0% to 35% depending on income brackets. VAT, an indirect tax, is levied on the value added at each stage of production and distribution, with a standard rate of 15% (Ghana Revenue Authority, 2023). Corporate tax, imposed on firms' profits, is set at 25%. Additional levies include the Capital Gains Tax, Customs and Excise Duties, and the National Health Insurance Levy (Ghana Revenue Authority, 2023). Ghana's tax framework aligns with international standards while accommodating domestic economic priorities. Government reliance on taxation shapes fiscal policy and determines the ability to provide essential services (Ogundana, Ibidunni, & Adetoyinbo, 2017; A. B. Olarinde & Arike, 2022). Income tax, VAT, and corporate tax constitute the core of government revenue, while non-tax sources

such as fees and fines play a smaller role. Maintaining an effective and sustainable tax system requires systematic governance and careful policy implementation.

Ghana has implemented numerous social welfare programs to mitigate socio-economic hardships and promote inclusive development. The Livelihood Empowerment Against Poverty (LEAP) program provides cash transfers to vulnerable households, including older adults, persons with disabilities, and orphans (Kipo-Sunyehzi, 2021; Ardianti & Sari, 2023). The National Health Insurance Scheme (NHIS) aims to achieve universal health coverage by improving access to medical services, while the Free Senior High School (SHS) policy reduces financial barriers to education. The Ghana School Feeding Program (GSFP) improves nutrition and school attendance among children (Tette & Enos, 2020). Collectively, these programs address poverty, enhance education and health outcomes, and contribute to social development.

These initiatives demonstrate Ghana's commitment to leveraging taxation and welfare programs to support socio-economic development and reduce inequality. By examining the interplay between taxation and welfare, this study seeks to provide empirical insights to guide evidence-based policy reforms and strengthen the country's efforts toward inclusive growth.

1.3 Statement of the Problem

Taxes are a fundamental instrument for economic development in Ghana, providing the government with the resources necessary to fund public services and stimulate growth (Eshun, 2018; Kooshkebaghi et al., 2022a). Ghana relies heavily on taxation to support critical programs, including healthcare, education, and poverty alleviation initiatives (Kipo-Sunyehzi, 2021; Tette & Enos, 2020). However, concerns have emerged regarding the extent to which these taxes influence the welfare of Ghanaians, particularly among vulnerable populations (Anyaduba & Otulugbu, 2019; Omesi & Appah, 2021). Like many developing economies, Ghana faces multi-faceted challenges in aligning taxation with social welfare objectives. The efficiency and equity of tax policies are critical determinants of socio-economic outcomes, with indirect taxes, such as value-added tax (VAT) and excise duties, raising questions about their disproportionate burden on low-income households (Andoh & Nkrumah, 2022; Andreyeva et al., 2022). Such regressive

effects may exacerbate inequality and impede efforts to reduce poverty (Piketty & Saez, 2007; Palić, Hodžić, & Dumičić, 2019). The substantial informal sector in Ghana further complicates the taxation–welfare nexus. Operating largely outside the formal tax system, this sector presents significant challenges to revenue mobilization and equitable distribution of tax burdens (Aringo, 2023; Anim, Awotwe, Nyarku, & Kusi, 2020). These difficulties are compounded by limitations in tax compliance, enforcement, and institutional capacity, which can undermine the effectiveness of taxation in promoting social welfare (Ardianti & Sari, 2023; Cantero-Galiano, 2022). Given these considerations, there is an urgent need to investigate the relationship between taxation and welfare in Ghana. This study seeks to examine the structure, progressivity, and distributional effects of the tax system across different income groups, while also evaluating the compliance challenges faced by taxpayers and the differential impact of taxes on various segments of society. Furthermore, it assesses the efficiency and effectiveness of welfare programs financed through taxation, considering factors such as governance, resource allocation, and infrastructure constraints (Jensen & Thompson, 2020; Franklin et al., 2022).

To achieve these objectives, the research adopts a quantitative approach, analyzing tax data from the Ghana Revenue Authority, World Development Indicators, and the Ghana Statistical Service (Ghana Revenue Authority, 2023; Abbam et al., 2018). The analysis examines trends, progressivity, and distributional impacts, alongside socio-economic indicators including income distribution, GDP growth, and access to education and healthcare (Li & Bai, 2022; Safarova, 2021; Morrisson, 2002). Ultimately, this study aims to generate empirical evidence on the effects of taxation on welfare outcomes in Ghana. The findings are expected to provide actionable recommendations for policymakers, focusing on tax reform, improved compliance, and the optimization of welfare programs. By elucidating the relationship between taxation and social welfare, this research contributes to evidence-based policy formulation that promotes inclusive and sustainable development.

1.4 Purpose of the Study

This research aims to investigate how taxation policies influence social welfare in Ghana. Taxation is intrinsically linked to economic, social, and political structures; understanding its effects

on welfare provides a foundation for developing more equitable and effective fiscal systems. The study seeks to generate insights that can inform policymakers, economists, and social scientists on the welfare implications of different tax policies, both at the community and national levels.

Specifically, the study examines the relationship between taxation and welfare by analyzing how various types of taxes impact access to healthcare, education, income distribution, and poverty reduction. It evaluates the effectiveness of different tax systems in promoting poverty alleviation, economic mobility, and individual investment. The research also considers the trade-offs between taxation and economic efficiency, exploring the extent to which citizens are willing to contribute for enhanced social welfare.

Furthermore, the study assesses taxation's role in advancing social equity by examining its effects on access to and quality of essential public services, including healthcare, education, and social protection mechanisms. Particular attention is given to disadvantaged groups, such as low-income households and marginalized communities, to evaluate how tax policy changes affect the distribution of welfare across different segments of society.

Based on these findings, the research will provide evidence-based policy recommendations aimed at maximizing the positive impact of taxation on social welfare. The recommendations will be contextualized for local and national settings and will highlight avenues for future research, thereby contributing to the development of tax systems that better serve societal welfare needs.

1.5 Research Objectives

1.5.1 General Objective

To examine the impact of taxation on welfare, focusing on its effects on income inequality, access to education, access to healthcare, and poverty reduction in Ghana.

1.5.2 Specific Objectives

1. To assess the effect of taxes on income inequality in Ghana.
2. To determine the effect of taxes on access to education in Ghana.
3. To determine the effect of taxes on access to healthcare in Ghana.
4. To assess the effect of taxes on poverty reduction in Ghana.

1.6 Research Questions

1. What is the effect of taxes on income inequality in Ghana?
2. What is the effect of taxes on access to education in Ghana?
3. What is the effect of taxes on access to healthcare in Ghana?
4. What is the effect of taxes on poverty reduction in Ghana?

1.7 Significance of the Study

The study of the impact of taxation on welfare in Ghana is of considerable importance from both policy and research perspectives. Understanding the interdependencies between taxation and social welfare is essential for evaluating how effectively taxes serve their intended purposes and assessing the socio-economic consequences of changes in tax policy. This study provides a systematic examination of how taxation influences income distribution, access to public services, and poverty alleviation, offering evidence to inform sound policy decisions.

First, the findings of this research can serve as an essential reference for Ghanaian policymakers in the formulation and implementation of tax policies that align more closely with national welfare objectives. By identifying the strengths and weaknesses of the existing tax structure, policymakers can design a system that is both equitable and efficient. Such reforms may involve

strategic allocation of public expenditures to areas where they are most needed, including social protection, education, healthcare, and poverty reduction programs that directly benefit the most vulnerable populations.

Second, the study's focus on the role of taxation in poverty alleviation provides insights into the mechanisms underlying income distribution and socio-economic inequality in Ghana. By identifying aspects of the tax system that may be regressive, the research enables interventions to mitigate adverse effects on low-income households, thereby promoting more just and inclusive economic growth. Evidence from this study can guide policies aimed at improving the quality of life and reducing economic disparities across different segments of the population.

Furthermore, investigating the linkages between taxation and the provision of public services, such as education and healthcare, can clarify the extent to which tax revenues are effectively translated into tangible benefits for citizens. This analysis supports evidence-based decisions regarding resource allocation, ensuring that public funds are directed toward high-quality, accessible services that strengthen human capital and foster sustainable socio-economic development.

In addition to its domestic relevance, the study contributes to the broader discourse on taxation and welfare in developing economies. Researchers, economists, and international development agencies can use the findings as a reference for comparative studies and cross-national policy analysis, facilitating the adoption of best practices and evidence-informed approaches.

Overall, this research aims to provide valuable insights for policymakers, supporting the development of taxation policies that promote inclusive and sustainable development in Ghana. By elucidating the relationship between taxation and welfare, the study seeks to inform strategies that advance equity, social justice, and economic prosperity for all citizens.

1.8 Delimitation of the Study

The scope of this study is confined to examining the direct relationship between taxation and social welfare in Ghana. Specifically, the research investigates how taxation policies influence citizens' welfare, focusing on government-imposed taxes, including income tax, consumption

tax, property tax, and social security contributions.

The study emphasizes the impact of taxation on key welfare dimensions, such as access to basic public services, income distribution, social protection, and overall economic development. It does not attempt to evaluate broader macroeconomic policies or other factors that may indirectly affect welfare outcomes over the long term, such as the quality of education, healthcare system efficiency, or external trade dynamics.

By identifying the specific linkages between taxation and welfare, this research aims to provide evidence-based insights that can guide policymakers in designing a more equitable and effective tax system to enhance citizen well-being.

1.9 Organization of the Study

This thesis is structured into six chapters, each addressing distinct aspects of the study.

Chapter One introduces the research, providing the background, purpose, and objectives of the study. It formulates the research questions, discusses the significance of the study, outlines delimitations, defines key terms, and presents an overview of the thesis structure. The chapter concludes with a summary of its main points.

Chapter Two presents a comprehensive review of the literature relevant to the study. This chapter discusses major concepts, theoretical perspectives, and empirical studies related to taxation and welfare. It also outlines the formulation of a conceptual framework that guides the research. The chapter ends with a synthesis of the reviewed literature and a summary of key findings.

Chapter Three describes the research methodology. It introduces the study area, research design, and the rationale for the chosen approach. The chapter provides detailed descriptions of the study population, sampling techniques, and data collection methods. It also explains the analytical procedures, ethical considerations, and limitations of the study, concluding with a summary of the methodological approach.

Chapter Four reports the research findings. It presents an introduction to the results, followed by detailed analyses of the collected data, response rates, and outcomes, referencing relevant tables and figures. The chapter concludes with a concise summary of the findings.

Chapter Five is dedicated to the discussion of the results. It interprets the findings in relation to the study objectives and contextualizes them within existing literature. Key discoveries are highlighted and critically analyzed to understand their implications for taxation and welfare in Ghana. The chapter ends with a summary of the discussion.

Chapter Six concludes the thesis by presenting the overall findings, drawing conclusions, and offering evidence-based recommendations. This chapter summarizes the study's contributions, policy implications, and potential directions for future research.

1.10 Chapter Summary

Chapter One of this thesis establishes the foundation for the study by providing a comprehensive background on the role of taxation as a fiscal policy instrument and its impact on social welfare. The chapter traces the evolution of taxation in Ghana, from traditional African societies through the colonial period and post-independence challenges, highlighting key transitions and reforms.

The problem statement clearly defines the primary objective of the study: to examine the relationship between taxation and social welfare by comparing the effects of progressive and regressive tax systems in Ghana and evaluating whether taxation policies can effectively promote social welfare. The chapter also articulates the research purpose, emphasizing the relevance of both policy formulation and scholarly investigation regarding taxation and welfare outcomes.

Research objectives and questions are outlined to guide the study, focusing on critical aspects such as income distribution, poverty alleviation, and potential reforms of the tax system. Furthermore, the significance of the study is discussed, illustrating its potential contributions to policymakers, resource allocation, and future research on taxation and welfare systems.

The delimitation statement specifies the scope of the study, clarifying that the focus is on taxation and welfare in Ghana, rather than broader social outcomes such as life expectancy

or general health indicators. Finally, the chapter provides an overview of the thesis structure, outlining the content and purpose of each subsequent chapter to guide the reader through the research process.

Chapter 2 Review of Literature

2.1 Introduction

This study focuses on the interplay between taxation and welfare in Ghana. This chapter provides a conceptual framework for understanding key ideas in the field and critically appraises relevant research. It presents a comprehensive review of the literature, highlighting the issues most pertinent to the study's research focus. All relevant findings, empirical studies, and reports from prior research have been carefully examined and analyzed to inform the study and establish its scholarly context.

2.2 Concept Review

This subsection reviews key concepts related to taxation and welfare within the context of the study area. For taxation, the review focuses on fundamental ideas such as the definition of taxation, direct and indirect taxes, income taxes, and consumption taxes. Regarding welfare, the discussion encompasses concepts such as welfare itself, income inequality, education, healthcare, and poverty reduction. The purpose of this review is to provide a conceptual foundation that guides the analysis of the relationship between taxation policies and social welfare in Ghana.

2.2.1 Concept of Taxation

Taxation refers to the revenue collected by state governments from individuals and businesses (Omesi & Appah, 2021). It serves as a mechanism for raising funds to compensate for externalities, stabilize the economy, and ensure equitable income distribution(Owuru & Olabisi, 2020). Anyaduba and Otulugbu (2019) define taxation as the collection of funds from people, enterprises, or other entities to finance public services and government expenditures, including income tax, sales tax, and property or corporate taxes. Similarly, (Palić et al., 2019) describe taxation as the government's levying of taxes on individuals and corporations, such as personal income tax, corporate profits, or declared turnover, at legally determined rates (Tadesse, 2023). The primary purpose of taxation is to expand public revenues and generate resources for government programs such as infrastructure development, environmental protection, healthcare, education, social welfare, and defense (Sahari et al., 2020; Kooshkebaghi, Emamgholipour, & Dargahi, 2022b). Taxation also functions as an instrument of wealth redistribution and economic stability. Progressive tax systems, in which higher earners pay a larger proportion of their income, enable the redistribution of wealth to lower-income groups (Ogundana et al., 2017; Ceyhan, Köstekçi, & Göv, 2021).

Moreover, taxation influences economic behavior and consumer preferences. Governments can use tax incentives or deductions to encourage specific activities, such as investment in renewable energy or research and development (Pokharel, 2018; Ntiamoah & Asare, 2020). However, while taxation is essential for financing public goods and services, it can also generate public debate regarding fairness, efficiency, and its impact on economic development (Fikr, Johari, Sholeh, & Suprayitno, 2021; Nemec, Pompura, & Šagát, 2015). Striking a balance between generating revenue and minimizing costs for taxpayers remains a persistent challenge for governments worldwide (Cantero-Galiano, 2022). In summary, taxation is a critical tool for revenue generation, social equity, and economic policy. Nevertheless, it continues to present challenges related to fairness, efficiency, and public acceptance, making it an enduring focus of research and policy deliberation.

2.2.2 Concept of Direct Taxation

Direct taxation is a fundamental component of fiscal policy and involves the imposition of taxes directly on individuals or entities by a government authority (Rahmat, Abd Hamid, Razak, & Sulaiman, 2023). Unlike indirect taxes, which are levied on goods and services and passed on to consumers, direct taxes are imposed directly on the income, wealth, or property of taxpayers (Kryeziu, 2021). Over time, the concept of direct taxation has evolved in response to changing economic structures and political ideologies (Tikhonova, Telegina, Babanskaya, & Grudneva, 2019). Historically, direct taxation primarily consisted of taxes on land, property, and income (Balamatsias, 2018). However, with economic modernization and diversification, the scope of direct taxation expanded to include taxes on personal income, corporate profits, capital gains, and inheritance. This evolution reflects governments' need to broaden revenue sources to finance public expenditures in increasingly complex economies.

Direct taxation is characterized by several key features. First, it promotes transparency because taxpayers are aware of the amount they pay and the basis upon which it is calculated (Yilmaz, Ozyer, & Ozyer, 2019). Second, many direct taxes are progressive, meaning that the tax rate increases as income or wealth increases. This progressivity helps ensure a more equitable distribution of the tax burden across income groups. Third, direct taxation serves as an important instrument for income redistribution and the reduction of socioeconomic inequalities. Scholars broadly agree that the defining feature of direct taxation is its direct relationship between the taxpayer and the government without intermediary transactions(Khoruzhy et al., 2020; Das Gupta & Gupta, 2017). This distinguishes direct taxes from indirect taxes, which are usually collected through intermediaries such as businesses.

In Ghana, direct taxation is governed by the Income Tax Act, 2015 (Act 896), which provides the legal framework for assessing and collecting taxes on individual and corporate income (Anim et al., 2020). The Act covers several forms of direct taxes, including personal income tax, corporate income tax, and capital gains tax. The Ghana Revenue Authority (GRA) is responsible for the administration and enforcement of these tax laws (Anim et al., 2020). In summary, direct taxation constitutes an essential element of public finance, involving the direct imposition of taxes on individuals and organizations based on income, wealth, or property. It is characterized

by transparency, progressivity, and a direct link between the taxpayer and the government.

2.2.3 Concept of Indirect Taxation

Indirect taxation refers to a form of taxation in which the burden of the tax is shifted from the entity that initially pays the tax to another party, typically the final consumer (Yilmaz et al., 2019). This occurs because indirect taxes are imposed on goods and services at the point of production or consumption rather than directly on individuals or businesses (Fredrick & Peter, 2019). Historically, indirect taxes have included levies such as excise duties, value-added taxes (VAT), and customs duties. These taxes are usually collected by intermediaries such as manufacturers, wholesalers, or retailers, who subsequently incorporate the tax into the price of goods and services. As a result, the final consumer ultimately bears the tax burden.

Over time, the scope of indirect taxation has expanded to include additional forms of regulatory taxes such as environmental taxes and so-called "sin taxes," which are imposed on products such as tobacco and alcohol. This evolution reflects the increasing complexity of modern economies and governments' efforts to raise revenue while simultaneously influencing consumer behavior. One of the main characteristics of indirect taxation is its regressive nature, as it often places a relatively higher burden on lower-income households compared to wealthier individuals. This occurs because low-income households typically spend a larger proportion of their income on consumption, which is subject to indirect taxation.

Researchers widely acknowledge that indirect taxes serve dual roles: generating government revenue and influencing market behavior (Lu, Zhao, Li, & Cao, 2023). By adjusting tax rates on specific products, governments can encourage or discourage particular types of consumption, thereby supporting broader policy objectives such as public health protection, environmental sustainability, or economic development.

In Ghana, indirect taxation is governed by several statutory frameworks. The Value Added Tax Act, 2013 (Act 870) regulates the administration of VAT, while the Excise Duty Act, 2014 (Act 878) provides for the imposition of excise duties on specific goods. Additionally, the Customs Act, 2015 (Act 891) governs the collection of customs duties on imported goods.

2.2.4 Concept of Income Tax

Income tax refers to a tax imposed on income earned by individuals and corporations within a specific tax period (Chumaidi & Noor, 2022). It represents one of the most significant sources of government revenue and is widely used to finance public expenditures and essential government services. Income taxation plays a critical role in modern fiscal systems by ensuring that individuals and corporations contribute a portion of their earnings to support national development. Governments rely on income taxes to fund public goods and services such as infrastructure, healthcare, education, and social protection programs. In addition to revenue generation, income taxation also contributes to economic stability and social equity by redistributing wealth and reducing income disparities.

Under most tax systems, individuals and businesses are required to declare their income and calculate the tax payable according to specified tax rates and regulations. These systems often incorporate progressive tax structures, in which higher income levels are taxed at higher rates. Such systems are designed to ensure fairness and enhance income redistribution within society. Furthermore, income taxation can influence economic behavior through tax incentives and deductions. Governments often introduce tax credits or exemptions to encourage investments in sectors such as renewable energy, research and development, and education. These incentives help stimulate economic growth while aligning individual economic activities with national development goals. Overall, income tax is a central instrument of fiscal policy, serving multiple purposes including revenue mobilization, redistribution of income, and the promotion of economic development.

2.2.5 Concept of Personal Income Tax

Personal income tax refers to a tax levied on income earned by individuals from various sources such as wages, salaries, interest, dividends, and rental income (Omesi & Appah, 2021). It is a form of direct taxation imposed by governments on individuals within their jurisdiction (Kato & Toyofuku, 2018). Personal income tax constitutes an important source of government revenue and plays a crucial role in financing public services and social programs. Historically,

income taxes were primarily imposed on wealthy individuals; however, modern tax systems apply personal income tax across multiple income brackets, thereby broadening the tax base.

Over time, the definition of personal income tax has expanded to include various forms of income beyond wages and salaries. These now include dividends, interest income, rental earnings, capital gains, and income from self-employment. Modern tax systems also incorporate deductions and tax credits to encourage socially desirable behaviors such as homeownership, retirement savings, and educational investment. One of the most important characteristics of personal income tax is its progressive structure, in which tax rates increase with rising income levels (Gobachew, Debela, & Shibiru, 2018). This progressive design helps promote equity in the tax system and contributes to reducing income inequality. Personal income tax therefore serves not only as a revenue-generating mechanism but also as a policy tool for addressing income disparities and promoting economic justice.

2.2.6 Gini Coefficient

The Gini coefficient is a widely used measure of inequality in income distribution, calculated based on the Lorenz curve (Omesi & Appah, 2021). The Lorenz curve, developed by Lorenz (1905), graphically represents the distribution of income or wealth in a society, showing the cumulative proportion of income held by cumulative population segments (Didenko, Skripnuk, & Kikkas, 2017; Fellman, 2018). The Gini coefficient quantifies deviation from perfect equality, with values ranging between 0 and 1: 0 indicates absolute equality, whereas values closer to 1 indicate higher income inequality (Hasell, 2023).

Income inequality can occur within a country or between countries. For instance, Ortiz and Cummins (2011) reported that in 2008, the average Gini coefficient in Sub-Saharan Africa was 0.442, compared to 0.483 for Latin America, 0.354 for Eastern Europe and Central Asia, and 0.309 for high-income countries. Other measures, such as decile or quintile ratios, are also used to evaluate income distribution. Nevertheless, the Gini coefficient remains the most popular and widely applied metric in empirical studies on income inequality (Hasell, 2023; Anyaduba & Otulugbu, 2019).

2.2.7 Concept of Access to Education

Access to education is a fundamental right and a cornerstone of social and economic development. It encompasses the availability, affordability, and inclusivity of educational opportunities for all individuals, irrespective of socioeconomic background, gender, or disability status. Access enables individuals to acquire knowledge, develop skills, and reach their full potential, thereby promoting social mobility and reducing inequality (UNESCO, 2021).

In Ghana, access to education is constitutionally recognized and supported by government policies. Initiatives such as the Free Senior High School (SHS) program have reduced financial barriers, enabling more students to enroll in secondary education (Government of Ghana, 2017). Programs addressing gender disparities, including the Ghana School Feeding Program and the Capitation Grant, have improved attendance and retention rates, particularly among girls from marginalized communities (Ministry of Education, 2019). Additionally, the Inclusive Education Policy seeks to accommodate children with disabilities within mainstream schools, ensuring equitable learning opportunities (Ministry of Education, 2019).

Despite these efforts, challenges persist, including inadequate infrastructure, limited educational resources, and regional disparities. Continuous investment and policy reforms are required to ensure that access to education in Ghana is inclusive, equitable, and of high quality.

2.2.8 Concept of Access to Healthcare

Access to healthcare refers to the ability of individuals and communities to obtain necessary medical services, including preventive, curative, and rehabilitative care, in a timely and affordable manner (World Health Organization, 2018). It involves factors such as physical proximity to healthcare facilities, financial resources, health insurance coverage, and cultural or language considerations (Andersen & Davidson, 2017; Penchansky & Thomas, 1981). Equitable access also depends on the quality, acceptability, and equity of healthcare provision.

In Ghana, the National Health Insurance Scheme (NHIS), implemented in 2003, represents a major policy intervention to reduce financial barriers to healthcare, particularly for marginalized

populations (Agyepong et al., 2016). Nevertheless, disparities persist, especially between urban and rural areas, due to challenges in transportation, infrastructure, and availability of skilled healthcare professionals (Gal & Bleikh, 2019). Cultural practices and beliefs also influence healthcare utilization, underscoring the importance of culturally sensitive interventions.

Quality differences further complicate healthcare access, with urban centers often offering more advanced facilities and specialized personnel compared to rural regions (Yawson, Aikins, & Antwi, 2019). Continued investment in healthcare infrastructure, workforce development, and policy implementation is essential to ensure equitable access across Ghana.

2.2.9 Concept of Poverty Reduction

Poverty reduction encompasses strategies aimed at alleviating and ultimately eradicating poverty within specific populations or geographic areas (Mbughi, 2018). It involves multidimensional interventions targeting income inequality, access to basic services, employment, and sustainable economic activities (World Bank, 2020; Ravallion, 2016). Effective poverty reduction strategies include income support programs, social protection mechanisms, access to education and healthcare, and community empowerment initiatives (United Nations Development Programme, 2018; D. Narayan, Chambers, Shah, & Petesch, 2000).

In Ghana, poverty reduction is a key policy priority, addressed through programs such as the Livelihood Empowerment Against Poverty (LEAP) program and the Ghana School Feeding Program (Ghana Ministry of Gender, Children and Social Protection, 2020; Ghana School Feeding Programme, 2020). Education policies, including the Free SHS initiative, aim to break intergenerational cycles of poverty by providing equitable learning opportunities for low-income households (Government of Ghana, 2017). Expansion of the NHIS and healthcare infrastructure further supports vulnerable populations (National Health Insurance Authority, 2020). Economic empowerment initiatives targeting agriculture, microfinance, and skills development also enable individuals and communities to generate income and achieve self-reliance (Ministry of Food and Agriculture, 2017). Nonetheless, regional disparities in access to education, healthcare, and economic opportunities remain significant (Ghana Statistical Service, 2018).

In conclusion, Ghana's poverty reduction strategy combines social safety nets, education and healthcare interventions, and sustainable economic empowerment programs. While progress has been achieved, continuous efforts are needed to address inequalities and enhance the effectiveness of these strategies.

2.3 Theoretical Review

This chapter presents the theoretical framework underpinning this study, with a focus on the multifaceted relationship between taxation and socio-economic outcomes, including inequality, education, healthcare, and poverty reduction. The theoretical discussion integrates classical and contemporary perspectives, encompassing the Endogenous Growth Model, Ability-to-Pay Theory, Human Capital Theory, Theory of Infrastructure-led Development, Benefit Theory, Theory of Realization and Exigent Implementation, and the social welfare concept. Together, these frameworks provide a comprehensive lens for examining how taxation affects public resource allocation, individual well-being, and societal development. The selected theories were chosen based on their explanatory power in addressing the distributive, economic, and developmental implications of taxation policies in contemporary contexts.

2.3.1 The Benefit Theory

The Benefit Theory of taxation proposes that taxpayers should contribute to government revenues proportionally to the benefits they receive from public services (Ayodeji & Others, 2021). This theory emphasizes an equity-efficiency trade-off, suggesting that the allocation of tax burdens can reflect both usage and derived advantages. For instance, those who gain substantial benefit from public infrastructure, healthcare, or security services are expected to contribute commensurately through taxation. This approach aligns with market-like mechanisms in public finance, where taxes serve as a price signal for public goods and help guide the government in resource allocation (Hermawan & Sinaga, 2020). Morimura (2019) conceptualizes proportional income tax as a form of benefit taxation, asserting that individuals' tax contributions should mirror their economic advantages from state protection. Peters further highlights that the so-

cial, legal, and economic frameworks established by governments facilitate income generation and security, creating a proportional link between individual benefits and tax obligations. Thus, the Benefit Theory supports policies that balance equity and efficiency while targeting income redistribution indirectly.

The practical application of the Benefit Theory is illustrated through excise taxes, such as the gasoline tax, where revenues are earmarked for public goods like road construction and maintenance. Consumers with higher consumption levels contribute more, reflecting greater derived benefits. Despite its theoretical appeal, the Benefit Theory encounters limitations in practice. Most public goods, including education and healthcare, are consumed collectively, making precise benefit measurement challenging. Furthermore, strict adherence may disproportionately impact low-income groups who, relative to their income, derive significant utility from public services. Consequently, modern tax policy often integrates the Benefit Theory with progressive taxation principles to achieve both efficiency and equity. In summary, the Benefit Theory provides a normative framework for linking taxation to service provision, illustrating how government revenue collection can reflect societal preferences while mitigating inequitable distributions.

2.3.2 Ability-to-Pay Theory

The Ability-to-Pay Theory posits that individuals should be taxed according to their financial capacity, typically measured by income, wealth, or consumption (Jhingan, 2014). This theory is grounded in principles of fairness and justice, advocating progressive taxation where higher earners contribute a larger share of public revenue. Its significance lies in the direct potential to reduce income inequality by redistributing resources to fund social services, infrastructure, and welfare programs. Tax systems based on the Ability-to-Pay principle also support fiscal sustainability by ensuring that those most able to contribute bear an appropriate share of the revenue burden. The theoretical rationale extends beyond income distribution; it underscores societal obligations to fund collective goods and services, such as public education, healthcare, and social protection programs, which benefit all citizens, regardless of whether they directly utilize these services.

Application of the Ability-to-Pay Theory in addressing income inequality involves progressive taxation schemes that balance fiscal responsibility with social justice. Critics, however, highlight limitations. Jhingan (2014) argues that reliance solely on consumption expenditure ignores wealth accumulation and investment, potentially misrepresenting taxpayers' true capacity. Appah (2019) notes that excessively progressive systems may discourage productivity and savings, while Omesi and Appah (2021) affirm that appropriately structured policies can promote welfare maximization without significant economic distortion. In practice, governments often combine this theory with other tax frameworks, including indirect taxes and benefit-based levies, to ensure both fairness and revenue efficiency. Overall, the Ability-to-Pay Theory remains central to contemporary tax policy design, emphasizing the link between income, capacity, and societal obligations, while serving as a mechanism for reducing economic disparities.

2.3.3 Human Capital Theory

Human Capital Theory emphasizes the economic value of investments in knowledge, skills, and health, asserting that such investments enhance individual productivity and societal prosperity (Zimmer, 2022; Jiang & Shi, 2023). Human capital encompasses formal education, vocational training, and health interventions, all of which collectively increase labor market productivity, wage potential, and innovation capacity (Yaqoob & Others, 2022). The theory asserts that governments and individuals should view expenditure on education and healthcare as investments with quantifiable economic returns. Policies that expand access to schooling, vocational programs, and healthcare services directly contribute to human capital accumulation, facilitating long-term economic growth and social mobility. Human Capital Theory also accounts for lifelong learning, emphasizing continuous skill development as essential for adapting to technological advances and labor market shifts (Rothomi & Rafid, 2023). Health, as a component of human capital, significantly influences productivity and economic participation; healthier populations are more capable of sustaining economic activities, contributing to social welfare (Kucharska, 2022).

Application of Human Capital Theory in taxation and education highlights the role of public investment in human development. Progressive taxation, informed by the Ability-to-Pay

principle, funds educational infrastructure, scholarships, and training programs, ensuring equitable access to opportunities (A. Olarinde & Arike, 2022). By financing human capital, governments enhance labor productivity, reduce income disparities, and stimulate sustainable economic growth. Critiques of the theory note its potential oversimplification of socio-economic inequalities and the complexity of skill acquisition (MacKenzie & Others, 2023; Yaqoob & Others, 2022). Nevertheless, Human Capital Theory remains foundational for understanding the role of taxation in promoting education, health, and workforce development, demonstrating the interdependence of fiscal policy and human development outcomes.

2.3.4 Theory of Infrastructure-led Development

The Theory of Infrastructure-led Development posits that strategic investments in foundational infrastructure are essential drivers of economic growth and societal transformation (Buthelezi, 2021). Infrastructure facilitates the efficient movement of goods, people, and information, enhancing productivity and enabling access to services such as education, healthcare, and commerce. Well-planned infrastructure reduces transaction costs, promotes business development, and catalyzes technological adoption. Within the context of taxation, revenue allocation to infrastructure development ensures that public resources are used to support productive investments that yield long-term societal benefits.

Application of this theory to taxation and education suggests that investments in school buildings, educational technology, and connectivity improve learning outcomes and human capital productivity. Infrastructure investments complement educational initiatives by creating environments conducive to learning and enhancing equitable access across socio-economic groups. Tax-funded infrastructure, aligned with strategic development goals, therefore becomes a pivotal mechanism for fostering economic growth, reducing inequality, and supporting sustainable human capital accumulation (A. Olarinde & Arike, 2022). By combining infrastructure-led development with progressive taxation, governments can simultaneously promote equity, efficiency, and long-term economic resilience.

2.3.5 Theory of Realization and Exigent Implementation

The Theory of Realization and Exigent Implementation asserts that taxation should yield tangible, timely, and beneficial outcomes for taxpayers without causing undue economic strain (O. Adegbite, 2021). This theory emphasizes the accountability of government in deploying revenues to deliver essential services, including healthcare, education, and public goods. Tax systems guided by this framework ensure that citizens perceive a direct linkage between contributions and public benefits, reinforcing compliance and trust.

In healthcare, this theory underscores that tax revenues should directly improve access, quality, and equity of medical services, particularly for marginalized populations. Ineffective utilization of tax revenue diminishes public confidence and undermines the potential of fiscal policies to generate social welfare. By ensuring the prompt and equitable allocation of resources, this theory bridges the gap between fiscal policy design and societal outcomes, highlighting the ethical and practical dimensions of taxation.

2.3.6 Social Welfare and Taxation for Poverty Reduction

The concept of social welfare provides a normative foundation for assessing taxation in relation to poverty reduction (Maniquet & Neumann, 2021). Maniquet and Neumann (2021) propose tax schemes that balance individual well-being with poverty alleviation, emphasizing that labor income taxation should minimize the effort required to achieve basic living standards. By integrating considerations of leisure, productivity, and subsistence, this framework aligns fiscal policy with social welfare objectives. Taxation under this model supports equitable redistribution, funding social protection, healthcare, education, and poverty reduction programs. While practical limitations exist, including labor market frictions and imperfect information, the social welfare framework remains a crucial guide for designing tax policies that enhance societal well-being and reduce economic inequality.

2.4 Empirical Review

2.4.1 Taxation and Income Inequality

Several empirical studies have investigated the relationship between taxation and income inequality, particularly in the context of developing economies like Nigeria. Omesi and Appah (2021) analyzed the impact of different tax types, including personal income tax, company income tax, petroleum profit tax, customs duties, excise duties, and value-added tax on the Gini coefficient as a measure of income inequality. Using empirical analysis and literature review, the study revealed that personal and company income taxes exhibit a significant negative relationship with income inequality, indicating that progressive taxation can contribute to reducing disparities. Conversely, petroleum profit tax was positively associated with income inequality, whereas value-added tax showed an insignificant positive correlation. The study highlighted that mismanagement of government spending on health and education can exacerbate inequality, emphasizing the need for effective allocation of revenue and enhanced tax compliance. Despite offering valuable insights, the study faced limitations, such as restricted geographic focus, limited discussion of potential data biases, and omission of certain socio-economic factors that may influence inequality.

Building on this, Xue (2018) conducted a literature-based analysis to assess the redistributive impact of direct taxes in Nigeria. Their findings indicated a strong positive relationship between revenue from personal income tax and petroleum profit tax and improved income redistribution, while company and education taxes had minimal impact. The study argued that taxation, particularly direct taxation, not only serves as a revenue-generating mechanism but also plays a central role in achieving equitable income distribution, thereby addressing structural imbalances. The authors emphasized the dual importance of robust tax administration and strategic allocation of resources to facilitate social welfare and infrastructure development.

In contrast, Obaretin et al. (2017) utilized a 34-year time series dataset to examine the influence of taxation on the Gini coefficient in Nigeria. The results suggested that direct, indirect, and total taxes did not significantly affect income inequality, indicating that the Nigerian tax sys-

tem may not adequately function as a redistributive tool. While the study employed rigorous statistical methods, including trend analysis, its reliance on secondary data, absence of detailed evaluation of individual tax types, and focus solely on Nigeria limited the generalizability of the findings.

Collectively, these studies demonstrate a nuanced and context-dependent relationship between taxation and income inequality. Progressive direct taxes, when effectively collected and combined with strategic allocation of public resources to education and healthcare, can significantly reduce income disparities. However, indirect taxes or poorly administered tax structures may fail to achieve this outcome, highlighting the importance of institutional capacity, governance quality, and complementary fiscal policies. Future research should explore cross-country comparisons, consider the impact of different tax instruments in varying institutional contexts, and incorporate comprehensive socio-economic variables to generate robust, policy-relevant insights.

2.4.2 Taxation and Education

Empirical research examining the relationship between taxation and education financing highlights the pivotal role of government revenue in supporting educational development. A. Olarinde and Arike (2022) investigated the impact of taxation on education financing in Nigeria using econometric approaches, including regression models, cointegration analysis, Vector Error Correction Models (VECM), and Granger causality Wald tests. Their study revealed a positive and statistically significant relationship between taxation and educational financing, demonstrating that increased tax revenue allows governments to improve education services, expand access, and enhance the quality of education. These findings corroborate earlier research by Babalola and Aminu (2011), who similarly highlighted taxation as a critical mechanism for mobilizing resources for educational development in resource-constrained settings.

Oluyebo (2022) analyzed the influence of petroleum profit tax and education tax on education outcomes and broader economic development in Nigeria. The study found a positive correlation between petroleum profit tax revenue and tertiary school enrollment, as well as life expectancy at birth, indicating that taxation can indirectly support both human capital formation and health outcomes. Interestingly, education tax revenue displayed a negative relationship with tertiary

enrollment, which may reflect inefficiencies in collection or allocation mechanisms. The study also observed that both taxation streams positively influenced per capita gross national income, highlighting the broader socio-economic impact of effective tax mobilization on educational and economic outcomes.

Ayangbah and Sun (2017) investigated the effect of taxation on higher education affordability in Nigeria, concluding that government grants should be increased to cover rising tertiary education costs. While the study offered practical policy recommendations, its reliance on qualitative assessment and absence of quantitative data limited the robustness and generalizability of its findings. Additionally, the exclusive focus on Nigeria restricts cross-context applicability, especially for countries with different fiscal structures, such as Ghana.

Taken together, these empirical studies underscore the crucial nexus between taxation and education financing. Effective tax administration, combined with progressive and well-targeted tax policies, can enhance government capacity to fund education, thereby fostering human capital development. However, challenges persist, including inefficiencies in tax collection, misallocation of funds, and limited cross-country comparative evidence. Future research should examine the differential effects of direct versus indirect taxes on educational outcomes, evaluate allocation efficiency, and explore the interplay between taxation, governance, and education policy to provide actionable insights for policymakers aiming to strengthen human capital through fiscal interventions.

2.4.3 Empirical Review on Taxation and Healthcare

The relationship between taxation and healthcare provision has received considerable attention in empirical research, particularly in developing economies where public health systems are heavily dependent on government revenues. (A. Adegbite, 2021) conducted a comprehensive study examining the relationship between taxation and health services in Nigeria, employing advanced econometric techniques to analyze both short-run and long-run dynamics. The study utilized cointegration analysis to explore the long-run equilibrium relationship between taxation and health services and employed a Vector Error Correction Model (VECM) to capture the short-run adjustments. Findings revealed a significant positive relationship between taxation

and healthcare provision, suggesting that higher tax revenues facilitate increased government expenditure on healthcare services. The study also confirmed earlier findings by Adisa-Atta (2017) in Ghana, reinforcing the notion that taxation can serve as a crucial instrument for improving public health systems. Notably, Adegbite emphasized the importance of a sustained tax base and effective revenue mobilization as critical drivers for healthcare financing. While the study is methodologically robust, it is limited by its focus on Nigeria, and the generalizability of its results to other contexts remains uncertain.

In Ghana, Adisa-Atta (2017) examined public attitudes toward higher taxation or user fees intended to improve government healthcare expenditure. Using cross-tabulation, correlation analysis, and multiple linear regression, the study explored whether willingness to pay higher taxes correlated with demographic factors, healthcare access, perceptions of government performance, and perceived corruption. The study found that only 35% of respondents supported higher taxes or user fees, despite widespread challenges in accessing quality healthcare. Regression results indicated a positive association between support for higher taxation and perceived improvement in healthcare delivery, highlighting the potential role of citizen acceptance in effective health financing. The study's strength lies in its use of rigorous quantitative techniques to identify determinants of public willingness to fund healthcare, though it did not assess the longitudinal impact of tax increases on actual health service delivery.

Boyd (2014) investigated the potential impact of alternative healthcare spending scenarios on state and local government budgets in the United States, highlighting the substantial growth of healthcare expenditures over the past 25 years. The study reported that state and local government healthcare spending nearly doubled relative to the overall economy, driven by increases in Medicaid and non-Medicaid expenditures. The study is notable for its extensive analysis of budgetary trends, providing a macro-level perspective on healthcare financing. However, it did not examine the specific contribution of tax revenue to this growth, leaving unanswered questions regarding the direct link between taxation and health expenditure.

Doherty (2014) explored the relationship between tax revenue and public healthcare financing in South Africa, combining quantitative and qualitative analyses. The study demonstrated that increases in tax revenue do not automatically translate into higher government health expenditure; instead, broader fiscal and economic policies, such as tax rate adjustments and

debt repayment strategies, play a significant role. This study contributes to understanding the complex policy environment influencing health financing. However, it lacks a detailed year-by-year analysis of how tax allocations impact healthcare service delivery, limiting the ability to draw conclusions about causal relationships.

In summary, empirical evidence indicates a generally positive association between taxation and healthcare provision, though outcomes are highly context-dependent. While studies in Nigeria and Ghana emphasize the potential of taxation to improve public health services, research in South Africa and the United States highlights the influence of broader fiscal policy decisions. These findings underscore the importance of not only raising tax revenue but also ensuring effective allocation and management of funds to maximize health service delivery. Collectively, these studies provide a foundation for understanding taxation as a policy tool for healthcare financing while revealing critical gaps for further empirical investigation, including longitudinal assessments and cross-country comparisons.

2.4.4 Empirical Review on Taxation and Poverty Reduction

Adu et al. (2023) conducted a study on the relationship between tax systems, poverty, and inequality. The authors first assume a nonlinear effect of tax systems on poverty and inequality, which is largely absent in the existing literature. They move beyond standard nonlinear treatments by testing whether the relationships are asymmetric. Additionally, the study considers both income and non-income measures of poverty to provide a more comprehensive assessment. Taxes were decomposed into direct and indirect categories, and the total tax effect was analyzed. Time-series data from 1983 to 2016 were gathered from multiple sources. The authors applied the novel NARDL (Nonlinear Autoregressive Distributed Lag) estimation approach while using ARDL as a baseline technique to examine the asymmetric effects of tax systems on poverty and inequality. Results reveal that tax systems exert asymmetric effects, with indirect taxes exerting a more significant impact on poverty and inequality than direct taxes. The study recommends that policymakers improve the progressivity of indirect taxes and reduce reliance on them to enhance the distributional impact of Ghana's tax system.

Lofgren and Cicowiez (2022) conducted a study titled "Taxation, infrastructure investment,

growth, and poverty reduction: A case study of Zimbabwe," providing a detailed analysis of the interplay between taxation policies, infrastructure investment, economic growth, and poverty reduction in Zimbabwe. Using macro- and meso-level Social Accounting Matrices (SAMs) and cross-country data from the World Bank, the authors examined Zimbabwe's economic performance from 2000 to 2014. They applied a Computable General Equilibrium (CGE) model to simulate the impact of different policy scenarios on key macroeconomic indicators, including GDP, investment, and poverty rates. Their findings indicate that increased infrastructure investment, particularly in energy and transport, could substantially enhance economic growth and reduce poverty. The study further emphasizes that reforms in taxation, such as broadening the tax base and reducing corporate income tax rates, could generate additional resources for infrastructure investment and social spending. These insights highlight the complex interdependencies among taxation, public investment, and poverty alleviation.

Lustig (2017) provides an extensive analysis of fiscal policy effects on inequality and poverty in 29 low- and middle-income countries. The study employs fiscal incidence methodologies to analyze the distributional consequences of government revenues and spending, including personal income taxes, payroll taxes, direct transfers, and consumption taxes. Household surveys, using income or consumption as welfare indicators, were applied. Findings suggest that fiscal policy can effectively reduce poverty and inequality; however, the impact depends on the size and composition of government revenues and expenditures. The author stresses the importance of investing in education and healthcare as mechanisms for inclusive growth and poverty reduction. These results offer critical insights into the capacity of fiscal policy to enhance equitable development outcomes.

Hirvonen et al. (2016) examined the effectiveness of taxation and social protection in Ethiopia. Key findings include the success of the Productive Safety Net Program (PSNP) in reducing poverty, with potential for expansion to currently excluded areas. The study also highlights the risk of income taxation becoming regressive if tax brackets are not regularly adjusted for inflation. Petersen (2008) assessed decentralization, integration, taxation, and revenue-sharing strategies, concluding that decentralization can enhance efficiency, transparency, and social welfare, although improper implementation may exacerbate inequalities. Leventi et al. (2017) studied tax-benefit policies in seven European countries, finding that child benefits, housing sup-

port, and social assistance reduce poverty, but effectiveness varies by country context. Kanbur et al. (2018) analyzed optimal taxation and public provision in developing countries, emphasizing linear tax policies, the role of informality, and the impact of public spending on well-being and poverty reduction.

Overall, these studies collectively underscore the multifaceted and context-dependent relationship between taxation and poverty reduction. While taxation can serve as a significant tool for promoting social welfare, outcomes are contingent upon tax structure, policy progressivity, and efficient allocation of public resources. The literature reveals a critical need for further empirical studies across different countries and longitudinal analyses to comprehensively understand taxation's role in poverty alleviation.

2.5 Research Gaps

A research gap represents an area within a field of study that has been insufficiently explored or inadequately addressed in existing literature. Identifying such gaps is critical, as they delineate the problem that the current research aims to address. Although extensive scholarly inquiries have examined the relationship between fiscal policy and various welfare indicators—such as income inequality, access to education, healthcare provision, and poverty alleviation—there remains a paucity of research focusing specifically on the Ghanaian context. In particular, few studies have comprehensively investigated the impact of taxation on income inequality, educational attainment, healthcare accessibility, and poverty reduction in Ghana. This lack of targeted research constitutes a notable gap in the literature and underscores the need for context-specific empirical analyses.

Moreover, the absence of up-to-date and reliable data on taxation and welfare outcomes in Ghana presents additional challenges for both researchers and policymakers. Existing studies often rely on outdated datasets, generalized cross-country comparisons, or aggregated measures that fail to capture the nuanced effects of different tax types on specific welfare indicators. For example, the differential impact of direct taxes (such as personal income tax, corporate tax, and capital gains tax) versus indirect taxes on income redistribution, educational financing,

or healthcare provision remains underexplored. Similarly, the indices and proxies commonly employed to measure welfare outcomes may not encompass all relevant dimensions, potentially overlooking critical aspects of well-being and socio-economic equity.

Consequently, this study seeks to address these gaps by undertaking a comprehensive empirical analysis of taxation in Ghana and its influence on key welfare indicators. By examining both direct and indirect taxes alongside measures of income inequality, educational outcomes, healthcare access, and poverty reduction, this research aims to provide robust, context-specific evidence to inform policy decisions. The study also emphasizes the need for improved data collection and methodological rigor to capture the full spectrum of taxation's socio-economic impacts, thereby contributing to more effective and equitable fiscal policy formulation.

Chapter 3 Research Methodology

3.1 Introduction

The central focus of this study is to analyze the impact of taxation on welfare in Ghana, as outlined in the preceding sections. This chapter details the research methodology employed to investigate the relationship between taxation policies and welfare in Ghana. It encompasses the research design and strategy, data sources, model specification, definition of variables, apriori expectations, estimation techniques, and data analysis procedures, and concludes with a summary of the chapter.

3.2 Research Design and Strategy

This study adopts a positivist research philosophy. Positivism emphasizes the importance of empirical evidence and seeks to explain phenomena through systematic observation and measurement (Noriey, 2022). In the context of this research, a positivist approach is essential for objectively assessing the impact of taxation on welfare, specifically its effects on welfare indicators such as income inequality, education, healthcare, and poverty reduction in Ghana, relying on quantifiable data and rigorous statistical analysis.

This research employs a quantitative research design, which is well-suited for analyzing numerical data to establish relationships and detect trends (Hadista, Suyadi, & Aisyah, 2021). Through quantitative analysis, this study aims to provide a precise and objective understanding of how

taxation policies influence welfare outcomes over a defined period.

The research strategy employed is Time Series Analysis. This approach involves collecting observations over successive time intervals, enabling the examination of correlations and trends across different income brackets and tax policies (Auer, Cavanaugh, Petor, Kinney, & Landers, 2022). This methodology is particularly effective for evaluating the temporal impact of taxation on income inequality. The time horizon for this study spans 38 years, from 1985 to 2022, allowing for a comprehensive assessment of taxation policies on income inequality within a relatively stable macroeconomic environment, while capturing variations resulting from policy changes and economic fluctuations.

3.3 Data Source

The study utilizes annual time-series data spanning from 1985 to 2022, yielding 38 observations, to examine the impact of taxation on welfare in Ghana. Time-series data refers to a sequence of observations collected over successive periods, which permits the analysis of temporal dynamics and trends in economic and social indicators over time. To ensure comprehensive coverage and accuracy, this study draws on multiple reliable data sources, including national and international statistical repositories. A primary source is the **Ghana Statistical Service (GSS) StatsBank**, which provides official aggregated time-series data on key indicators such as national accounts, demographic characteristics, education statistics, health outcomes, poverty measures, and labor and income statistics for Ghana. StatsBank allows users to build customized tables and download historical datasets necessary for longitudinal analysis (Ghana Statistical Service, 2023). In addition, the **World Bank World Development Indicators (WDI)** database offers standardized time-series data covering measures such as GDP per capita, poverty headcount ratios, Gini coefficients, foreign direct investment inflows, and education and health outcomes, facilitating international comparability and econometric modeling over extended periods (World Bank, 2023). Furthermore, specific welfare-related statistics, such as demographic and health indicators, are available through the **Ghana Demographic and Health Survey (GDHS)**, which provides national time-series data on health access, mortality, fertility, and service utilization, useful for analyzing changes in healthcare access over time (Ghana Demographic and

Health Survey (GDHS), 2023). For income inequality measures, the Gini index for Ghana is available from datasets compiled by the **World Bank and its partner databases**, which aggregate annual series on inequality indicators derived from household surveys and national sources (Federal Reserve Bank of St. Louis (FRED), 2023).

3.4 Model Specification

This study employs a multiple regression analysis framework to investigate the relationship between taxation (independent variables) and welfare indicators (dependent variables), including income inequality, access to education, access to healthcare, and poverty reduction, while controlling for relevant covariates. Four models were specified to achieve the study objectives.

3.4.1 The First Model: Taxation and Income Inequality

Income inequality is expressed as a function of taxation:

$$\text{Income Inequality} = f(\text{Taxation}) \tag{3.1}$$

The functional model is:

$$\text{GINI} = f(\text{Taxation}) \tag{3.2}$$

$$\text{GINI} = \mu_0 + \mu_1 \text{TDT} + \mu_2 \text{TTR} + \mu_3 \text{TIT} + \mu_4 \text{OPI} + \mu_5 \text{FDI} + \mu_6 \text{INF} + \epsilon \tag{3.3}$$

Where:

- GINI = Gini Coefficient
- μ_0 = Intercept term

- TDT = Total Direct Tax Revenue
- TTR = Total Tax Revenue
- TIT = Total Indirect Tax Revenue
- OPI = Economic Openness
- FDI = Foreign Direct Investment
- INF = Inflation Rate
- ϵ = Error term

3.4.2 The Second Model: Taxation and Education

The relationship between taxation and education is specified as:

$$\text{EDUCATION} = f(\text{TREV}, \text{NONREV}, \text{PCGDPR}, U_2) \tag{3.4}$$

Where access to education is proxied by Secondary School Education Enrollment (SSEE):

$$\text{SSEE} = b_0 + b_1\text{TREV} + b_2\text{NONTREV} + b_3\text{PCGDPR} + u_2 \tag{3.5}$$

Where:

- SSEE = Secondary School Education Enrollment
- TREV, NONTREV, PCGDPR = Taxation and economic control variables
- u_2 = Error term
- b_0 = Constant term
- b_1–b_3 = Coefficients of explanatory variables

3.4.3 The Third Model: Taxation and Access to Healthcare

The model is expressed as:

$$\text{Healthcare Access} = f(\text{Tax System}) + Z \tag{3.6}$$

Where:

- Tax System = Direct Tax, Indirect Tax, Total Tax
- Z = Non-tax revenue, Inflation Rate
- HCA = Healthcare Access

3.4.4 The Fourth Model: Taxation and Poverty Reduction

The growth-poverty relationship is expressed as:

$$P_i = \alpha_0 + \beta_1 \ln \text{GDPPC}_t + \epsilon_t \tag{3.7}$$

Where:

- P_i = Poverty indicators (headcount, gap, squared)
- GDPPC_t = Real GDP per capita
- ϵ_t = Error term
- α_0 = Intercept
- β_1 = Percentage change in poverty associated with income

Extending the model to include tax systems and control variables:

$$P_i = \alpha_0 + \beta_2 \ln \text{taxsys}_i + \beta_1 \ln \text{GDPPC}_t + \beta_3 \ln \text{GI} + \beta_4 \ln \text{GEX} + \beta_5 \ln \text{GP} + \beta_6 \ln \text{AG} + \epsilon_t \quad (3.8)$$

Where:

- taxsys_i = Tax systems (Direct, Indirect, Total)
- GI = Gini Index
- GEX = Government Expenditure
- GP = Gold Price
- AG = Agriculture Value Added
- ln = Natural logarithm
- ϵ_t = Error term

3.5 Definition of Variables

To ensure clarity and consistency in the analysis, the key variables employed in this study are defined and summarized in Table 3.1. Each variable represents a specific aspect of taxation or welfare outcomes, allowing for precise measurement and interpretation in the regression models. The definitions provided facilitate understanding of the relationships examined in the study and support the operationalization of the variables within the quantitative framework.

3.5.1 Control Variables Justification and Poverty Measures

In the regression models specified, several control variables are incorporated to account for other factors that could influence welfare outcomes in Ghana, ensuring the estimated effects of taxation are not biased. In the income inequality model, economic openness (OPN) is included

Table 3.1: Definition of Variables Used in the Study

Variable	**Definition**
Income Inequality (Gini Coefficient)	Unequal distribution of income among households.
Education	Acquisition of knowledge and skills through formal or informal means.
Healthcare	Prevention, treatment, and management of diseases and promotion of overall well-being.
Poverty Reduction	Efforts to decrease poverty and improve living conditions.
VAT	Consumption tax levied on goods and services.
Customs and Excise Duties	Taxes on imported or exported goods.
Personal Income Tax	Tax on individual income.
Company Tax	Tax on corporate profits.
Capital Gains Tax	Tax on profits from asset sales.
Error Term	Captures variability not explained by the model.

to account for the potential impact of trade and foreign market integration on income distribution, as open economies can influence labor and capital income across sectors. Foreign Direct Investment (FDI) is included because inflows of capital can create employment opportunities and influence wage structures, potentially affecting inequality. Inflation (INF) serves as a control for price level changes that could erode real income and affect disposable income, indirectly influencing income distribution. These control variables allow the model to isolate the effects of taxation types (direct, indirect, and total) on income inequality while accounting for broader economic factors.

For the education model, control variables such as non-tax revenue (NONTREV) and per capita GDP (PCGDPR) are included to capture alternative funding sources and overall economic capacity for education. These factors are relevant because governments often rely on multiple revenue streams, and higher per capita income can improve household affordability for education, thereby influencing enrollment rates. Similarly, in the healthcare model, non-tax revenue and inflation are incorporated to control for fiscal capacity beyond taxation and the effects of rising prices on health service accessibility. This ensures that the observed impact of taxation on healthcare access reflects policy effects rather than broader economic fluctuations.

In the poverty reduction model, several control variables are also included to better understand the distributional and structural determinants of poverty. The Gini index (GI) is used to account for inequality, recognizing that higher income inequality may hinder poverty reduction even in

the presence of higher GDP per capita. Government expenditure (GEX) is included to capture the impact of fiscal spending on social programs that can alleviate poverty. Gold price (GP) serves as a proxy for commodity-dependent revenues in Ghana, which can influence national income and poverty outcomes, while agriculture value added (AG) reflects the contribution of agriculture to livelihoods, particularly in rural areas where poverty is concentrated. These controls help to isolate the specific effect of taxation on poverty, ensuring a more accurate interpretation of policy impacts.

To more comprehensively measure poverty reduction, this study incorporates both poverty headcount ratio and poverty intensity measures (poverty gap index and squared poverty gap). The headcount ratio captures the proportion of the population living below the poverty line, while the intensity measures reflect the depth and severity of poverty, accounting for how far below the poverty line households are. Including these complementary measures provides a fuller understanding of poverty dynamics and allows for a nuanced assessment of how direct, indirect, and total taxes influence poverty reduction in Ghana.

By incorporating relevant control variables and multiple poverty measures, this study ensures that the relationships between taxation and welfare indicators are robust, theoretically grounded, and empirically credible. These enhancements strengthen the alignment between the research design, models, and objectives, providing a solid foundation for policy-relevant conclusions.

3.6 Apriori Expectations

Based on theoretical and empirical literature, the following expectations are formulated for this study. Indirect taxes, such as value-added tax (VAT), customs duties, and excise duties, are expected to have a **positive relationship with income inequality** due to their regressive nature, which disproportionately affects lower-income households (μ_1–μ_2). Conversely, direct taxes, including personal income tax, corporate tax, and capital gains tax, are anticipated to exhibit a **negative relationship with income inequality**, reflecting their redistributive effect on income.

Regarding education, taxes are expected to positively impact access and enrollment, as fiscal revenues provide resources for public education programs (β_1–β_5). For healthcare, the expectation is similarly **positive**, as taxation enables government spending on hospitals, clinics, and medical infrastructure, thereby improving population health outcomes (α_1–α_5).

In terms of poverty reduction, indirect taxes are predicted to increase poverty incidence slightly due to their regressive structure, while direct taxes are anticipated to reduce poverty by providing revenue for social protection and welfare programs (δ_1–δ_5).

Additionally, the study incorporates **control variables** such as government expenditure, economic openness, foreign direct investment, and inflation, to capture other factors that may influence welfare outcomes. Including both poverty **headcount** and **intensity measures** ensures a comprehensive analysis, capturing not only the proportion of people living below the poverty line but also the depth of their deprivation, aligning the model closely with the study's objectives.

These expectations serve as a benchmark for evaluating empirical results, guiding interpretation of the relationship between taxation and key welfare indicators.

3.7 Estimation Technique, Strategy, and Procedure

The estimation followed four steps: (1) testing each variable for a unit root to establish its order of integration; (2) testing for a long-run (cointegrating) relationship using the Autoregressive Distributed Lag (ARDL) bounds testing approach; (3) estimating the short-run and long-run effects of taxation on each welfare indicator; and (4) checking the estimated models with diagnostic and stability tests. Each step is described below.

3.7.1 Unit Root Test

Macroeconomic time series such as tax revenue and GDP per capita usually trend over time. Regressing one trending (non-stationary) series on another can produce a high R^2 and significant

coefficients even when the variables are unrelated, a problem known as spurious regression. To avoid this, the stationarity of each variable was examined using the Augmented Dickey-Fuller (ADF) test (Dickey & Fuller, 1979), based on the regression:

$$\Delta y_t = \alpha + \delta t + \gamma y_{t-1} + \sum_{i=1}^{k} \beta_i \Delta y_{t-i} + \varepsilon_t \tag{3.9}$$

where $\Delta y_t = y_t - y_{t-1}$ is the first difference of the series, t is a time trend, k is the number of lagged differences included to remove serial correlation in the error term, and ε_t is the error term. The null hypothesis is $H_0 : \gamma = 0$ (the series has a unit root and is non-stationary) against the alternative $H_1 : \gamma < 0$ (the series is stationary). The test was conducted first at levels and then at first difference. A variable that is stationary at levels is integrated of order zero, I(0); a variable that becomes stationary only after first differencing is integrated of order one, I(1). Establishing the order of integration matters because the ARDL bounds test is valid for a mixture of I(0) and I(1) variables but not for variables that are I(2).

3.7.2 The ARDL Model

The study adopted the ARDL approach developed by Pesaran, Shin, and Smith (2001). The approach was preferred over the Johansen cointegration and Vector Error Correction Model (VECM) framework for three reasons. First, it can be applied whether the regressors are I(0), I(1), or a mixture of both, which suits the mixed results of the unit root tests. Second, it performs well in small samples, which is important given the 38 annual observations available for this study. Third, a single equation provides both the short-run dynamics and the long-run relationship for each welfare indicator. The general ARDL$(p, q_1, \ldots, q_k)$ model is specified as:

$$y_t = \alpha_0 + \sum_{i=1}^{p} \phi_i \, y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j} \theta_{jl} \, x_{j,t-l} + \varepsilon_t \tag{3.10}$$

where y_t is the welfare indicator (GINI, SSER, HCA, or POVH), $x_{j,t}$ are the tax and control variables specified in Section 3.4, p is the number of lags of the dependent variable, q_j is the number of lags of regressor j, and ε_t is the error term. The coefficients on the current values of

the regressors (θ_{j0}) capture the short-run effects, while the coefficients on their lags ($\theta_{j1}, \theta_{j2}, \ldots$) capture the lagged effects. Given the annual frequency of the data and the small sample, a maximum of two lags was allowed for each variable, and the optimal lag structure was selected using the Akaike Information Criterion (AIC). The models were estimated by Ordinary Least Squares (OLS).

3.7.3 Bounds Test for Cointegration

Cointegration means that variables which individually trend over time move together in a stable long-run relationship. To test for cointegration, the ARDL model is rewritten in the following error correction form:

$$\Delta y_t = \alpha_0 + \lambda_0 \, y_{t-1} + \sum_{j=1}^{k} \lambda_j \, x_{j,t-1} + \sum_{i=1}^{p-1} \phi_i^* \, \Delta y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j-1} \theta_{jl}^* \, \Delta x_{j,t-l} + \varepsilon_t \tag{3.11}$$

The null hypothesis of no long-run relationship, $H_0 : \lambda_0 = \lambda_1 = \cdots = \lambda_k = 0$, is tested against the alternative that at least one λ is non-zero using an F-test. The computed F-statistic is compared with two sets of critical values: a lower bound, which assumes all regressors are I(0), and an upper bound, which assumes all regressors are I(1). If the F-statistic exceeds the upper bound, the null hypothesis is rejected and cointegration is confirmed; if it falls below the lower bound, there is no cointegration; and if it lies between the two bounds, the test is inconclusive. Because the sample is small, the critical values of P. K. Narayan (2005), which are computed for samples of 30 to 80 observations, are more appropriate than the asymptotic values of Pesaran et al. (2001).

3.7.4 Long-Run and Short-Run (Error Correction) Estimates

Where cointegration is confirmed, the long-run coefficient of each regressor is obtained from the error correction form as:

$$\beta_j = -\frac{\lambda_j}{\lambda_0} \tag{3.12}$$

which measures the total effect of a permanent one-unit change in x_j on y after all adjustments have taken place. The short-run dynamics are then estimated using the Error Correction Model (ECM):

$$\Delta y_t = \alpha_0 + \sum_{i=1}^{p-1} \phi_i^* \, \Delta y_{t-i} + \sum_{j=1}^{k} \sum_{l=0}^{q_j-1} \theta_{jl}^* \, \Delta x_{j,t-l} + \psi \, ECT_{t-1} + \varepsilon_t \tag{3.13}$$

where ECT_{t-1} is the error correction term, the lagged deviation of y from its long-run equilibrium. Its coefficient, ψ, is the speed of adjustment: it is expected to be negative and statistically significant, and it shows the proportion of any deviation from the long-run equilibrium that is corrected within one year.

3.7.5 Diagnostic and Stability Tests

To confirm that the estimated models are reliable, the residuals of each model were subjected to diagnostic tests. The Breusch-Godfrey Lagrange Multiplier (LM) test (Breusch, 1978; Godfrey, 1978) was used to test for serial correlation, the Breusch-Pagan test (Breusch & Pagan, 1979) for heteroskedasticity, and the Jarque-Bera test (Jarque & Bera, 1980) for normality of the residuals. In each case, the null hypothesis is that the assumption holds (no serial correlation, constant variance, and normally distributed residuals), so a p-value above 0.05 indicates that the model passes the test. Finally, the stability of the estimated coefficients over the sample period was examined using the cumulative sum (CUSUM) test (Brown, Durbin, & Evans, 1975); the coefficients are judged stable if the CUSUM plot remains within the 5% significance bounds.

3.8 Data Analysis and Techniques

The data analysis proceeded in stages. First, descriptive statistics (mean, standard deviation, median, minimum, maximum, skewness, and kurtosis) were computed to summarise the characteristics of the variables over the 1985–2022 period. Second, a correlation matrix, trend plots, and scatter plots were used to explore the behaviour of the key variables over time and the preliminary relationships between them. Third, the ADF unit root test was applied to each variable at levels and at first difference. Fourth, separate ARDL models were estimated for each of the four welfare indicators, following the specific objectives of the study: income inequality (Objective 1), access to education (Objective 2), access to healthcare (Objective 3), and poverty reduction (Objective 4). For each model, the short-run, lagged, and long-run effects of taxation were obtained. Finally, the diagnostic and stability tests described above were conducted to validate the models before the results were interpreted.

3.8.1 Chapter Summary

This chapter presented the research methodology employed to examine the impact of taxation on welfare in Ghana. A positivist research philosophy guided the study, underpinned by a quantitative research design using annual time series data spanning 1985 to 2022 (38 observations). Data were drawn from the Ghana Statistical Service, the World Bank World Development Indicators, and the Ghana Revenue Authority, covering taxation variables, welfare indicators, and economic control variables. Four models were specified to investigate the relationships between taxation and income inequality, education, healthcare, and poverty reduction, with clear definitions of all key variables and apriori expectations grounded in theory and empirical literature. The estimation procedure involved the Augmented Dickey-Fuller unit root test, the ARDL bounds testing approach to cointegration, the estimation of short-run and long-run effects using the ARDL and error correction framework, and diagnostic and stability tests. This methodological framework provides the foundation for the empirical results presented in the next chapter.

Chapter 4 Results

4.1 Introduction

This chapter presents the empirical results of the investigation into the relationship between taxation and welfare outcomes in Ghana. The analysis systematically addresses each research objective by examining how tax structures influence income inequality, educational access, healthcare provision, and poverty reduction. Descriptive statistics provide an overview of the data characteristics, followed by trend analysis of key variables and unit root tests to establish stationarity properties. The chapter subsequently presents ARDL model estimations, exploring both short-run dynamics and long-run equilibrium relationships, concluding with diagnostic tests to validate the robustness of the findings.

4.1.1 Descriptive Statistics

Table 4.1 presents the summary statistics for the key variables employed in the study, covering the period 1985 to 2022. The mean Gini coefficient of 40.59 indicates moderate income inequality in Ghana, with relatively low variability (SD = 2.85) suggesting stability in income distribution over the study period. Poverty headcount (POVH) exhibits substantial variation (range = 76.33%), reflecting significant fluctuations in poverty levels, with a mean of 44.67% of the population living below the poverty line. GDP per capita averaged $1,231.57, demonstrating considerable economic dispersion (SD = 420.98) as the economy evolved. Secondary school enrollment (SSER) averaged 44.87%, showing progressive improvement from 29.83% to 75.31% by the end of the period. Government expenditure (GEX) averaged 9.79% of GDP with

moderate stability. Notably, inflation (INF) displays extreme volatility (SD = 20.57) and high kurtosis (10.94), confirming the presence of periods of macroeconomic instability, including the hyperinflationary episode reflected in the maximum value of 122.87%. These statistics provide essential context for understanding the welfare landscape within which Ghana's tax policies operate.

Table 4.1: Descriptive Statistics for Selected Variables

Variable	Mean	SD	Median	Min	Max	Range	Skew	Kurtosis
GINI	40.59	2.85	41.84	35.30	43.50	8.20	-0.71	-0.93
POVH	44.67	20.37	49.40	4.45	80.78	76.33	-0.14	-1.17
GDPPC	1231.57	420.98	1062.86	702.42	2046.01	1343.58	0.63	-1.11
SSER	44.87	14.54	37.43	29.83	75.31	45.48	0.86	-0.89
GEX	9.79	2.33	9.60	5.86	15.31	9.45	0.47	-0.65
INF	23.64	20.57	17.30	4.87	122.87	118.01	2.91	10.94

Figure 4.1 presents the pairwise correlation coefficients among the key variables, revealing important preliminary relationships in the data. The correlation matrix indicates a strong negative association between GDP per capita and poverty headcount ($r \approx -0.85$), consistent with the growth-poverty linkage theory, suggesting that economic expansion is associated with poverty reduction. Secondary school enrollment exhibits a moderate negative correlation with income inequality ($r \approx -0.47$), aligning with human capital theory, which posits that educational expansion promotes more equitable income distribution. Government expenditure demonstrates a negative correlation with poverty ($r \approx -0.52$), supporting the fiscal intervention hypothesis that public spending contributes to welfare improvement. Inflation displays positive correlations with both poverty and inequality, reflecting the regressive impact of macroeconomic instability on vulnerable households. The tax variables show complex relationships with welfare indicators, with direct taxes exhibiting slight negative correlations with inequality, consistent with their progressive structure, while indirect taxes demonstrate weakly positive correlations with poverty measures. These preliminary correlations provide foundational support for the subsequent econometric analysis, though they do not establish causal relationships.

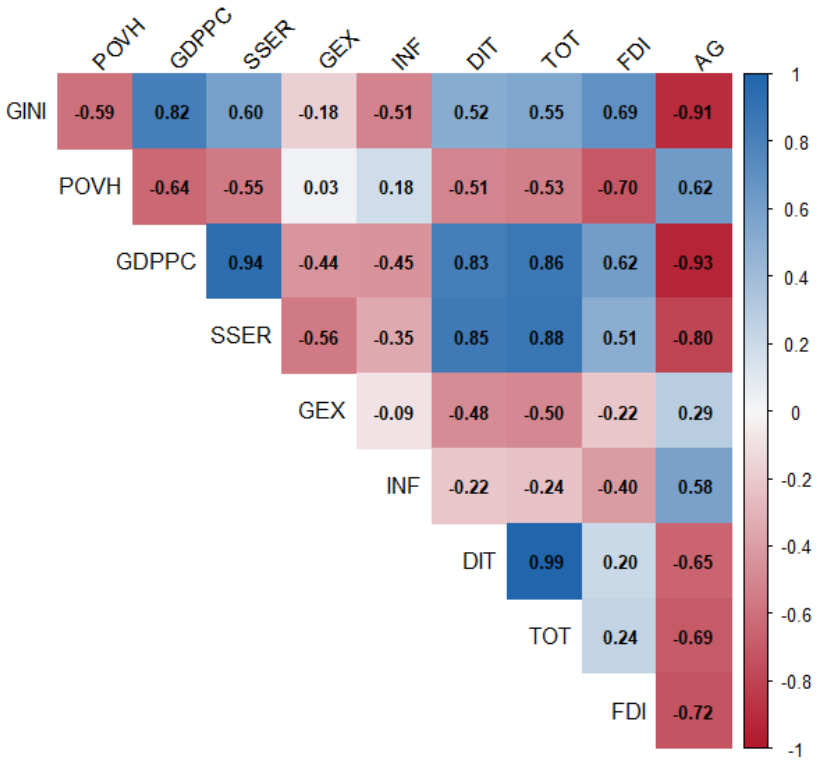


Figure 4.1: Correlation Matrix of Selected Economic and Social Variables

4.2 Trend Analysis of Key Variables

Figure 4.2 illustrates the temporal dynamics of secondary school enrollment (SSER), government expenditure (GEX), and inflation (INF) in Ghana from 1985 to 2022. Secondary school enrollment exhibits a pronounced upward trajectory, increasing from approximately 30% in the early 2000s to over 75% by 2022. This monotonic increase reflects the cumulative impact of educational policy interventions, including the Free Compulsory Universal Basic Education (FCUBE) program and the more recent Free Senior High School policy, which progressively expanded access to secondary education. Government expenditure as a percentage of GDP displays moderate fluctuation around the mean of 9.79%, with notable peaks corresponding to election years and periods of fiscal expansion. The relative stability of GEX suggests consistent public sector commitment to welfare-related spending, though the magnitude of fluctuation increased post-2010. Inflation demonstrates extreme volatility, characterized by sharp spikes in 2001 (following currency depreciation), 2009 (post-global financial crisis), and most dramatically in 2022, where it exceeded 120%. These inflationary episodes represent significant macroeco-

nomic shocks that erode household purchasing power and potentially offset welfare gains from educational expansion and government spending. The juxtaposition of rising educational attainment against inflationary instability provides critical context for understanding the complex welfare landscape in Ghana.

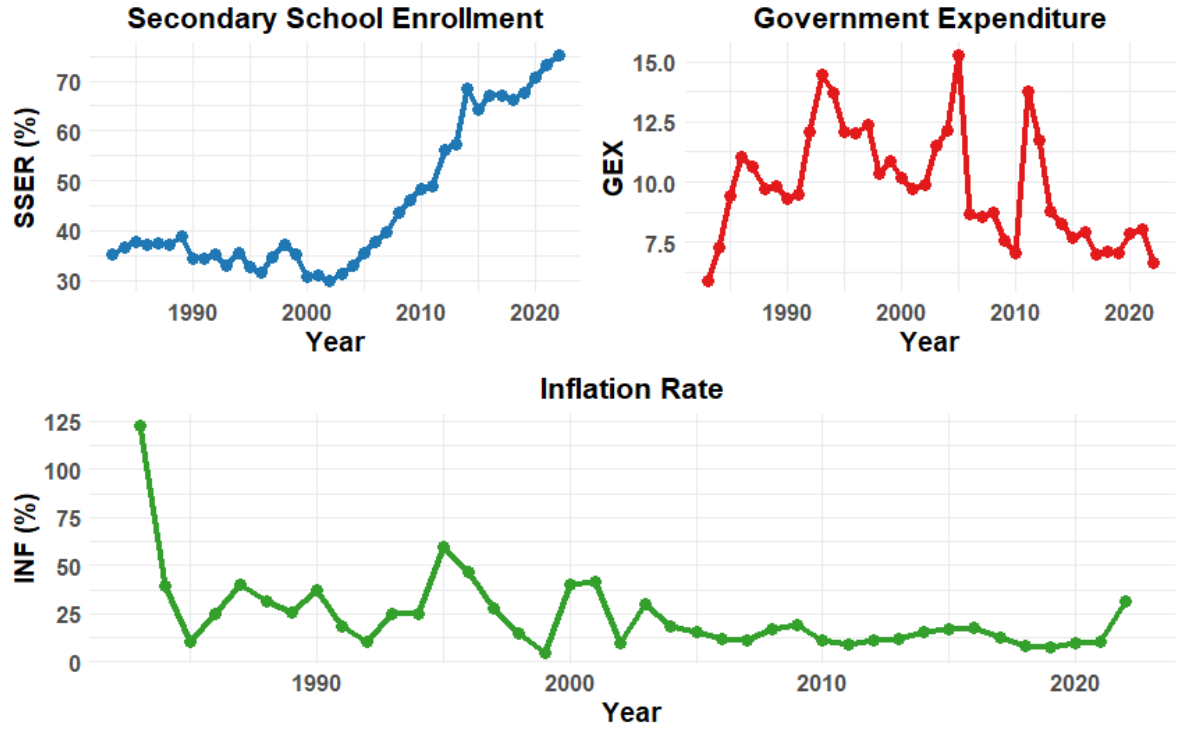


Figure 4.2: Time Series of Education, Government Expenditure, and Inflation Rate

Figure 4.3 depicts the evolution of income inequality (GINI), poverty headcount (POVH), and GDP per capita (GDPPC) in Ghana over the 1985–2022 period. GDP per capita exhibits a consistent upward trajectory, increasing from approximately $700 at the start of the sample period to over $2,000 by 2022, reflecting sustained economic growth driven by macroeconomic stability, commodity exports, and service sector expansion. This growth pattern aligns with Ghana's transition to lower-middle-income status in 2010. Poverty headcount demonstrates a pronounced secular decline, falling from approximately 80% in the early 2000s to below 5% by 2022. This dramatic reduction corresponds with the growth-poverty elasticity literature, indicating that Ghana's growth episode has been broadly pro-poor, supported by social protection programs including the Livelihood Empowerment Against Poverty (LEAP) initiative and the National Health Insurance Scheme (NHIS). Income inequality, measured by the Gini coefficient, displays a more complex pattern. Following a modest decline in the early 2000s,

inequality increased steadily from 2006 onwards, peaking around 2017 before exhibiting a slight moderation. The divergence between declining poverty and rising inequality post-2010 suggests that while growth has lifted households above the poverty line, the benefits have been unevenly distributed, with higher-income groups capturing a disproportionate share of economic gains—a pattern consistent with the Kuznets hypothesis in middle-income transitioning economies. This simultaneous reduction in poverty and increase in inequality presents a critical welfare paradox with significant implications for tax policy design.

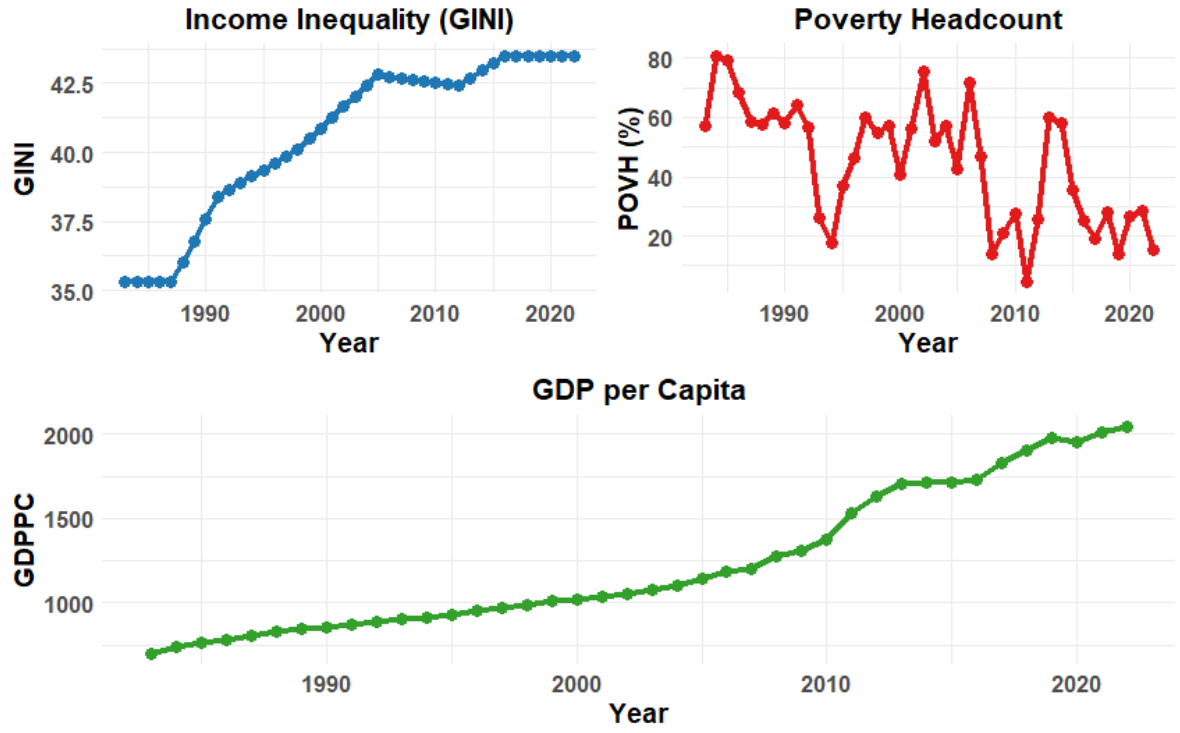


Figure 4.3: Time Series of Income Inequality, Poverty Headcount, and GDP per Capita

4.3 Exploratory Relationship Analysis

Figure 4.4 reveals a strong negative nonlinear relationship between GDP per capita and poverty headcount. As real incomes increase, poverty declines precipitously initially, followed by a more gradual reduction at higher income levels. This pattern conforms to the growth-poverty elasticity literature, suggesting diminishing marginal poverty reduction returns to economic growth. The tight clustering of observations around the fitted line indicates that economic expansion has been consistently associated with poverty reduction throughout the study period, though the

relationship appears to weaken at higher income thresholds. The plot displays a weak negative relationship between secondary school enrollment and the Gini coefficient, with considerable dispersion around the fitted line. This suggests that while educational expansion may contribute to more equitable income distribution—consistent with human capital theory—the relationship is moderated by other structural factors, including labor market segmentation and returns to education across different skill levels. The substantial scatter indicates that education alone cannot fully explain inequality dynamics. A moderate negative association emerges between government expenditure and poverty, supporting the fiscal intervention hypothesis. However, the relationship exhibits notable heterogeneity, with similar expenditure levels associated with varying poverty outcomes. This dispersion likely reflects differences in expenditure composition, efficiency of public service delivery, and targeting effectiveness across different periods. The plot reveals a weak positive association between inflation and poverty, though the relationship is heavily influenced by extreme observations, particularly the 2022 hyperinflationary episode. The general pattern suggests that macroeconomic instability imposes welfare costs on poor households, consistent with the regressive impact of inflation on real incomes and purchasing power.

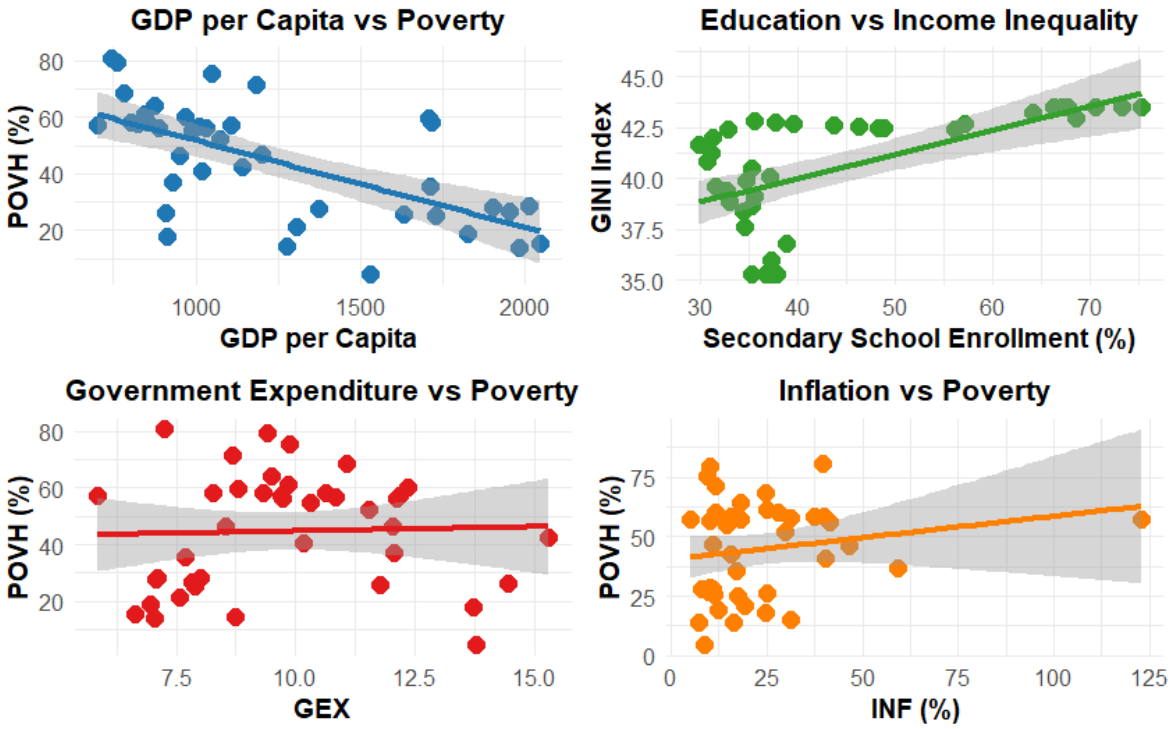


Figure 4.4: Scatter Plots Depicting Key Socioeconomic Relationships: GDP per Capita vs Poverty, Secondary School Enrollment vs Income Inequality, Government Expenditure vs Poverty, and Inflation Rate vs Poverty.

4.4 Unit Root Test Results

Before estimating the ARDL models, the stationarity properties of all variables were examined using the Augmented Dickey-Fuller (ADF) test, with the null hypothesis that the series contains a unit root (non-stationary). The test was first conducted on the variables at levels (Table 4.2) and then on their first differences (Table 4.3).

Table 4.2 presents the ADF test results at levels. **Stationary Variables at Level [I(0)]:** Inflation (INF), poverty headcount (POVH), government expenditure (GEX), and agriculture value added (AG) are stationary at the 5% significance level. These variables exhibit mean-reverting behavior, implying that shocks to these series are temporary and they return to their long-run equilibrium paths. The stationarity of INF ($p = 0.010$) confirms that inflationary episodes in Ghana, while volatile, do not permanently alter the inflation trajectory. The remaining variables—the Gini coefficient (GINI), direct taxes (DIT), indirect taxes (IDT), total

taxes (TOT), foreign direct investment (FDI), secondary school enrollment (SSER), and GDP per capita (GDPPC)—have p-values above 0.05 at levels, so the null hypothesis of a unit root cannot be rejected. These variables were therefore differenced once and tested again.

Table 4.2: ADF Unit Root Test Results at Levels

Variable	ADF Statistic	p-value	Decision
GINI	-1.8543	0.6307	Non-stationary
DIT	4.2997	0.9900	Non-stationary
IDT	1.2857	0.9900	Non-stationary
TOT	3.1880	0.9900	Non-stationary
FDI	-1.6214	0.7221	Non-stationary
INF	-5.8527	0.0100	Stationary, I(0)
SSER	-0.9136	0.9387	Non-stationary
POVH	-4.0564	0.0176	Stationary, I(0)
GDPPC	-1.3830	0.8156	Non-stationary
GEX	-3.8316	0.0277	Stationary, I(0)
AG	-4.1794	0.0128	Stationary, I(0)

Note: H_0: the series has a unit root. Decision based on the 5% significance level.

Table 4.3 presents the ADF test results at first difference. **Integrated of Order One [I(1)]:** Foreign direct investment (FDI) and secondary school enrollment (SSER) become stationary after first differencing at the 5% level. These variables contain stochastic trends, meaning shocks have permanent effects unless corrected through cointegrating relationships. The non-stationarity of SSER reflects the cumulative nature of educational attainment, while FDI's characteristics align with capital flow persistence. **Potential I(2) Processes:** The Gini coefficient (GINI), direct taxes (DIT), indirect taxes (IDT), total taxes (TOT), and GDP per capita (GDPPC) fail to achieve stationarity even after first differencing at conventional significance levels. The p-values remain above 0.05, suggesting these series may require second differencing or exhibit structural breaks. This finding has important implications: income inequality and tax variables display strong persistence, indicating that redistributive policies may have enduring effects. The borderline case of GINI ($p = 0.0565$ at first difference) suggests near I(1) behavior, warranting cautious interpretation. The mixed integration order—I(0), I(1), and potential I(2) variables—justifies the application of the ARDL bounds testing approach to cointegration, which accommodates variables with different stationarity properties and remains valid when the dependent variable is I(1) or I(0), provided no variable is I(2). Given the potential I(2) characteristics of some tax variables, robustness checks are essential to ensure valid inference.

Table 4.3: ADF Unit Root Test Results at First Difference

Variable	ADF Statistic	p-value	Order of Integration
ΔGINI	-3.5040	0.0565	May be I(2)
ΔDIT	-0.6885	0.9625	May be I(2)
ΔIDT	-2.6394	0.3230	May be I(2)
ΔTOT	-0.1298	0.9900	May be I(2)
ΔFDI	-4.2587	0.0101	I(1)
ΔINF	-7.5730	0.0100	I(0)
ΔSSER	-3.8340	0.0279	I(1)
ΔPOVH	-6.0120	0.0100	I(0)
ΔGDPPC	-3.5397	0.0512	May be I(2)
ΔGEX	-5.6684	0.0100	I(0)
ΔAG	-6.9440	0.0100	I(0)

Note: H_0: the series has a unit root. Δ denotes the first difference.

4.5 ARDL Model Estimation

This section presents the Autoregressive Distributed Lag (ARDL) model estimations examining the dynamic relationships between taxation and welfare outcomes in Ghana. The ARDL bounds testing approach is particularly appropriate given the mixed integration order of the variables, as shown in Tables 4.2 and 4.3, and its suitability for small sample sizes. Following the specific objectives of the study, the results are organised into four subsections: taxation and income inequality (Objective 1), taxation and access to education (Objective 2), taxation and access to healthcare (Objective 3), and taxation and poverty reduction (Objective 4). Under each objective, the short-run effects are presented first, followed by the lagged effects and the long-run relationship. A summary of the long-run equilibrium values for all the models is presented in Section 4.5.5.

4.5.1 Taxation and Income Inequality

The first objective of the study was to assess the effect of taxes on income inequality in Ghana. This subsection presents the ARDL estimates for the income inequality model, in which income inequality is measured by the Gini coefficient (GINI).

4.5.1.1 Short-Run Effects of Taxation on Income Inequality

Table 4.4 presents the short-run ARDL model estimates examining the contemporaneous effects of tax structure on income inequality, measured by the Gini coefficient. The model demonstrates strong explanatory power, with an adjusted R^2 of 0.9953 and a highly significant F-statistic (558.2, $p < 0.001$), indicating that the selected regressors collectively explain approximately 99.5% of the variation in income inequality. Both direct taxes (DIT) and indirect taxes (IDT) exhibit statistically insignificant effects on income inequality in the short run, with p-values of 0.5290 and 0.8461, respectively. The coefficients are economically negligible, with direct taxes showing a minute negative coefficient (-0.000021) and indirect taxes a positive coefficient (0.000009). This insignificance suggests that tax structure adjustments do not produce immediate distributional consequences, consistent with the existence of implementation lags and delayed transmission mechanisms in fiscal policy. Foreign direct investment (FDI) displays a negative but statistically insignificant coefficient (-0.0129, $p = 0.6678$), failing to support the hypothesis that capital inflows influence income distribution in the short term. Inflation (INF) similarly shows an insignificant negative relationship (-0.000689, $p = 0.8377$), indicating that price level changes do not contemporaneously affect inequality measures. The absence of significant short-run effects for all explanatory variables, despite the model's high explanatory power, suggests that income inequality in Ghana is predominantly determined by its own inertial dynamics rather than immediate responses to fiscal or macroeconomic shocks. This finding underscores the importance of examining lagged effects, presented in Table 4.5, to capture the delayed impact of tax policies on distributional outcomes. The constant term (1.2870, $p = 0.1381$) is also insignificant, confirming that the model specification adequately captures the systematic determinants of inequality without omitted variable bias.

Table 4.4: Short-term Effects of Tax Structure on Income Inequality

Variable	**Coefficient**	**Std. Error**	**p-value**
Intercept	1.2870	0.8375	0.1381
DIT	-0.000021	0.000033	0.5290
IDT	0.000009	0.000046	0.8461
FDI	-0.0129	0.0297	0.6678
INF	-0.000689	0.003325	0.8377
Observations		38	
R^2		0.9971	
Adjusted R^2		0.9953	
F-statistic		558.2 ($p < 0.001$)	

4.5.1.2 Lagged Effects of Taxation on Income Inequality

Table 4.5 presents the ARDL model estimates examining the lagged effects of tax structure and control variables on income inequality in Ghana, capturing the dynamic adjustment processes following changes in fiscal policy. The first and second lags of the Gini coefficient are highly significant, with coefficients of 1.5910 ($p < 0.001$) and -0.6199 ($p = 0.0010$), respectively. The positive coefficient on the first lag indicates strong persistence in income inequality—approximately 159% of an inequality shock persists into the subsequent period. The negative coefficient on the second lag, however, suggests a partial correction mechanism, where approximately 62% of the shock is subsequently reversed. This pattern implies that inequality disturbances in Ghana exhibit oscillatory convergence toward equilibrium, consistent with a second-order autoregressive process where shocks generate initial overshooting followed by partial adjustment. Neither direct taxes (DIT) nor indirect taxes (IDT) demonstrate statistically significant lagged effects on income inequality at either one or two periods. All coefficients carry p-values exceeding 0.59, with economically negligible magnitudes. The first lag of DIT shows a positive coefficient (0.000046), while the second lag turns negative (-0.000008), though both remain insignificant. Similarly, IDT coefficients are negative in both lags (-0.000010 and -0.000012) but statistically indistinguishable from zero. Foreign direct investment (FDI) and inflation (INF) also fail to achieve statistical significance at any lag. FDI exhibits unstable signs across lags (positive at lag one, negative at lag two), while inflation shows alternating signs (positive at lag one, negative at lag two) with consistently high p-values. The overwhelming insignificance of all explanatory variables—tax and control alike—contrasted with the strong significance of

the lagged dependent variables, indicates that income inequality in Ghana is predominantly an inertial phenomenon. The results suggest that distributional outcomes are primarily driven by historical inequality patterns rather than contemporaneous or lagged fiscal interventions. This finding poses important questions for tax policy effectiveness: if tax structures do not significantly influence inequality even with lags, the redistributive capacity of Ghana's fiscal system may be limited by structural factors including the large informal sector, tax evasion, or inefficient allocation of tax revenues. Alternatively, the results may reflect the relatively small magnitude of tax adjustments relative to the scale of underlying inequality dynamics, or the possibility that tax effects are absorbed by other macroeconomic variables not captured in the specification. The diagnostic properties of this model, as reported in subsequent tables, validate the reliability of these findings.

Table 4.5: Lagged Effects of Tax Structure on Income Inequality

Variable	**Coefficient**	**Std. Error**	**p-value**
$GINI_{t-1}$	1.5910	0.1601	0.0000***
$GINI_{t-2}$	-0.6199	0.1643	0.0010***
DIT_{t-1}	0.000046	0.000084	0.5917
DIT_{t-2}	-0.000008	0.000058	0.8910
IDT_{t-1}	-0.000010	0.000106	0.9249
IDT_{t-2}	-0.000012	0.000104	0.9120
FDI_{t-1}	0.0018	0.0360	0.9595
FDI_{t-2}	0.0108	0.0266	0.6881
INF_{t-1}	0.003211	0.003150	0.3186
INF_{t-2}	-0.002655	0.002013	0.2001

4.5.1.3 Long-Run Effect of Taxation on Income Inequality

The long-run equilibrium values for income inequality range from -3.608 to 7.594, with a mean of -0.101. The near-zero mean suggests that, in the long run, tax structure variables do not exert a systematically positive or negative effect on income inequality. The wide range, however, indicates considerable variation in the long-run relationship depending on model specification and lag structure. The negative mean, though small, hints at a redistributive effect of the overall tax system when all long-run adjustments are complete. However, the inclusion of zero within the plausible range of values suggests that this redistributive effect may not be statistically distinguishable from zero, consistent with the insignificant tax coefficients observed

in the short-run and lagged inequality models (Tables 4.4 and 4.5). This finding implies that Ghana's tax system may lack the progressivity or scale necessary to fundamentally alter the long-run distribution of income, possibly due to the large informal sector, limited direct tax coverage, or regressive indirect tax composition.

4.5.2 Taxation and Access to Education

The second objective of the study was to determine the effect of taxes on access to education in Ghana. Access to education is proxied by secondary school enrollment (SSER).

4.5.2.1 Short-Run Effects of Taxation on Access to Education

Table 4.6 presents the ARDL model estimates examining the contemporaneous relationship between direct taxation and secondary school enrollment in Ghana, incorporating GDP per capita and government expenditure as control variables. The model demonstrates strong explanatory power, with an adjusted R^2 of 0.9736 and a highly significant F-statistic (171.6, $p < 0.001$), indicating that the regressors collectively explain approximately 97.4% of the variation in secondary school enrollment. The coefficient for direct taxes is positive (0.000550) but statistically insignificant at conventional levels ($p = 0.2657$). While the positive sign aligns with theoretical expectations that tax revenues finance educational expenditures—consistent with human capital theory and the fiscal foundations of education policy—the lack of statistical significance suggests that contemporaneous changes in direct tax collections do not immediately translate into measurable improvements in secondary school enrollment. This finding is consistent with the existence of fiscal implementation lags: tax revenues must undergo appropriation, budget allocation, and disbursement before affecting educational outcomes, a process that typically spans multiple fiscal cycles. Additionally, the relatively small magnitude of the coefficient indicates that the short-run elasticity of educational enrollment with respect to tax revenue is economically modest. GDP per capita (GDPPC) exhibits a negative coefficient (-0.012404, $p = 0.4527$) that is statistically insignificant, failing to support the hypothesis that higher income levels directly increase enrollment rates in the short run. This counterintuitive sign may reflect the complex relationship between economic growth and educational investment decisions at the

household level, where income effects may be offset by rising opportunity costs of schooling during economic expansions—a phenomenon documented in the labor-education trade-off literature. Government expenditure (GEX) similarly shows an unexpected negative coefficient (-0.077562, p = 0.7545) that is statistically insignificant, suggesting that short-term variations in aggregate government spending do not contemporaneously affect enrollment outcomes. This may reflect the composition of government expenditure, where education-specific allocations may be diluted within broader fiscal aggregates, or inefficiencies in public expenditure management that delay the translation of spending into educational outcomes. The absence of significant short-run effects for all explanatory variables, despite the model's high explanatory power, indicates that secondary school enrollment in Ghana is predominantly driven by its own inertial dynamics and longer-term structural factors rather than immediate responses to fiscal variables. The insignificance of direct taxes in the short run is theoretically plausible, as educational outcomes respond slowly to revenue availability due to infrastructure development timelines, teacher recruitment and training cycles, and household decision-making processes that exhibit considerable persistence. These findings underscore the importance of examining lagged effects, presented in Table 4.7, to capture the delayed impact of tax policy on educational attainment. The constant term (0.92154, p = 0.8488) is also insignificant, confirming appropriate model specification and the absence of systematic omitted variable bias.

Table 4.6: Short-term Effects of Direct Taxes on Secondary School Enrollment

Variable	**Coefficient**	**Std. Error**	**p-value**
Intercept	0.9215	4.7893	0.8488
DIT	0.000550	0.000485	0.2657
GDPPC	-0.012404	0.016295	0.4527
GEX	-0.077562	0.245735	0.7545
Observations		38	
R^2		0.9793	
Adjusted R^2		0.9736	
F-statistic		171.6 ($p < 0.001$)	

4.5.2.2 Lagged Effects of Taxation on Access to Education

Table 4.7 presents the ARDL model estimates examining the lagged effects of direct taxation on secondary school enrollment in Ghana, capturing the dynamic adjustment processes following

changes in fiscal policy. The model includes lagged dependent variables and lagged explanatory variables to account for temporal dependencies in educational outcomes. The first lag of secondary school enrollment ($SSER_{t-1}$) is highly significant with a coefficient of 0.8104 ($p < 0.001$), indicating strong persistence in educational attainment. Approximately 81% of the previous period's enrollment level carries forward to the current period, reflecting the cumulative nature of educational progression—students enrolled in one period naturally continue to subsequent periods, and educational infrastructure investments exhibit considerable durability. This strong autoregressive component confirms that secondary school enrollment is inherently a slow-adjusting process. Unlike the insignificant contemporaneous effects in Table 4.6, the lagged tax variables achieve statistical significance, revealing delayed policy impacts. The first lag of direct taxes (DIT_{t-1}) carries a negative coefficient (-0.002468, p = 0.0102), while the second lag (DIT_{t-2}) shows a positive coefficient (0.002223, p = 0.0047). This alternating sign pattern suggests an overshooting-adjustment dynamic in the tax-enrollment relationship. The negative first lag may reflect short-term adjustment costs—increased tax collection could temporarily reduce household disposable income, potentially creating liquidity constraints that delay educational enrollment decisions. The positive second lag, however, indicates that after this adjustment period, the revenue-financed educational investments materialize, ultimately increasing enrollment. This pattern is theoretically consistent with the "fiscal lag" hypothesis, where tax increases initially impose private costs before generating public benefits. GDP per capita similarly exhibits significant lagged effects with alternating signs. The first lag ($GDPPC_{t-1}$) shows a positive coefficient (0.076079, p = 0.0062), while the second lag ($GDPPC_{t-2}$) displays a negative coefficient (-0.057429, p = 0.0021). The positive first lag aligns with income theory—higher economic growth increases household resources available for educational investment. The negative second lag may reflect labor market dynamics, where sustained growth eventually increases opportunity costs of schooling as employment prospects improve, potentially drawing students into the workforce rather than continued education. The significant lagged tax effects, contrasted with insignificant contemporaneous effects, confirm that fiscal policy transmission to educational outcomes operates with substantial delays. The two-year adjustment pattern suggests that tax revenues require approximately two fiscal cycles to be appropriated, allocated to educational sectors, and translated into tangible enrollment improvements. This finding has important policy implications: tax-based educational financing strategies require sustained

commitment, as immediate enrollment responses should not be anticipated. The results also highlight the importance of revenue stability—volatile tax collections could disrupt the positive second-year effects, undermining educational planning and infrastructure development. The strong persistence in enrollment further implies that policy interventions, once effective, generate durable improvements in educational access through cumulative progression effects.

Table 4.7: Lagged Effects of Direct Taxes on Secondary School Enrollment

Variable	**Coefficient**	**Std. Error**	**p-value**
$SSER_{t-1}$	0.8104	0.0923	0.0000***
DIT_{t-1}	-0.002468	0.000898	0.0102**
DIT_{t-2}	0.002223	0.000725	0.0047***
$GDPPC_{t-1}$	0.076079	0.025795	0.0062***
$GDPPC_{t-2}$	-0.057429	0.016954	0.0021***

4.5.2.3 Long-Run Effect of Taxation on Access to Education

The long-run equilibrium values for education outcomes range from 20.745 to 126.545, with a mean of 40.722. The positive mean indicates that, in the long run, tax revenues and associated fiscal variables exert a positive effect on secondary school enrollment. The magnitude suggests that sustained tax-financed educational investments ultimately expand access to secondary education, consistent with human capital theory and the fiscal foundations of education policy. The wide range reflects sensitivity to control variables and lag specifications, but the consistently positive central tendency confirms that taxation contributes to long-run educational development in Ghana. This finding aligns with the significant lagged tax effects observed in Table 4.7, where direct taxes positively influenced enrollment after a two-year adjustment period, confirming that the long-run educational benefits of taxation materialize through sustained revenue mobilization and allocation.

4.5.3 Taxation and Access to Healthcare

The third objective of the study was to determine the effect of taxes on access to healthcare in Ghana. Healthcare access (HCA) is modelled as a function of direct, indirect, and total taxes, with non-tax revenue and inflation as control variables, as specified in Chapter Three.

4.5.3.1 Long-Run Effect of Taxation on Access to Healthcare

The long-run equilibrium values for healthcare access range from 71.623 to 151.939, with a mean of 97.176. The substantial positive mean indicates that taxation exerts a strong positive effect on healthcare access in the long run, with the magnitude suggesting that sustained tax-financed health investments substantially expand population coverage and service availability. This finding supports the welfare state hypothesis—tax revenues enable governments to finance health infrastructure, medical personnel, and insurance schemes that ultimately improve healthcare access. The relatively high mean value (97.176) implies that the long-run elasticity of healthcare access with respect to tax revenue is economically meaningful, underscoring the importance of sustained revenue mobilization for health sector development. The range, while wide, remains consistently positive, confirming the robustness of this long-run relationship across alternative specifications.

4.5.4 Taxation and Poverty Reduction

The fourth objective of the study was to assess the effect of taxes on poverty reduction in Ghana. Poverty is measured by the poverty headcount ratio (POVH). Because the poverty model specified in Chapter Three builds on the growth–poverty relationship, this subsection also reports the effect of taxation on GDP per capita, the growth channel through which taxation may influence poverty reduction.

4.5.4.1 Short-Run Effects of Taxation on Poverty Reduction

Table 4.8 presents the ARDL model estimates examining the contemporaneous relationship between tax structure, control variables, and poverty headcount in Ghana. The model demonstrates moderate explanatory power, with an adjusted R^2 of 0.6174 and a statistically significant F-statistic (4.512, p = 0.0009), indicating that the regressors collectively explain approximately 61.7% of the variation in poverty headcount. The coefficient for direct taxes is negative (-0.001720) but statistically insignificant (p = 0.5308). While the negative sign aligns with theo-

retical expectations that direct taxation—particularly progressive income taxes—provides fiscal resources for poverty-reducing expenditures, the lack of statistical significance suggests that contemporaneous changes in direct tax collections do not immediately translate into measurable poverty reduction. This finding is consistent with the fiscal transmission lag hypothesis, where tax revenues require time to be appropriated and channeled into poverty alleviation programs before affecting household welfare. Government expenditure (GEX) emerges as the only statistically significant determinant of poverty headcount in the short run, with a negative coefficient of -4.32008 ($p = 0.0061$). This finding strongly supports the fiscal intervention hypothesis—increases in government spending contemporaneously reduce poverty, reflecting the direct impact of social protection programs, public service delivery, and infrastructure investments on household welfare. The magnitude of the coefficient suggests that a one percentage point increase in government expenditure as a share of GDP is associated with approximately a 4.32 percentage point reduction in the poverty headcount ratio, an economically meaningful effect. GDP per capita (GDPPC) exhibits a negative coefficient (-0.04873, $p = 0.6599$) that is statistically insignificant, failing to support the growth-poverty linkage hypothesis in the short run. This counterintuitive finding may reflect the distributed lag structure of growth effects—economic expansion requires time to generate employment and income gains that lift households above the poverty line. Income inequality (GINI) shows a positive but insignificant coefficient (5.72918, $p = 0.7138$), while agriculture value added (AG) displays a positive insignificant coefficient (0.63874, $p = 0.6504$), suggesting that short-term fluctuations in these structural variables do not contemporaneously affect poverty outcomes. The dominance of government expenditure as the sole significant poverty determinant in the short run has important policy implications. Unlike tax revenues, which must undergo appropriation processes before affecting welfare, government spending directly injects resources into the economy through social transfers, public employment, and service provision, generating immediate poverty reduction effects. This finding underscores the distinction between revenue mobilization and expenditure effectiveness in poverty alleviation strategy—tax policy alone does not reduce poverty; rather, it is the utilization of tax revenues through well-targeted government spending that generates welfare improvements. The insignificance of direct taxes in the short run reinforces the importance of examining lagged effects, presented in Table 4.9, to capture delayed poverty reduction impacts following tax policy changes. The constant term (-33.9117, $p = 0.8590$) is statistically

insignificant, confirming appropriate model specification.

Table 4.8: Short-term Effects of Tax Structure and Controls on Poverty Headcount

Variable	**Coefficient**	**Std. Error**	**p-value**
Intercept	-33.9117	188.3966	0.8590
DIT	-0.00172	0.00270	0.5308
GDPPC	-0.04873	0.10911	0.6599
GINI	5.72918	15.40176	0.7138
GEX	-4.32008	1.41005	0.0061***
AG	0.63874	1.38828	0.6504
Observations		38	
R^2		0.7932	
Adjusted R^2		0.6174	
F-statistic		4.512 (p = 0.000893)	

4.5.4.2 Lagged Effects of Taxation on Poverty Reduction

Table 4.9 presents the ARDL model estimates examining the lagged effects of tax structure and control variables on poverty headcount in Ghana, capturing the dynamic adjustment processes following changes in fiscal policy. The model includes lagged dependent variables and lagged explanatory variables to account for temporal dependencies in poverty dynamics. Both the first and second lags of poverty headcount are statistically significant, with coefficients of 0.79395 (p = 0.0005) and -0.42941 (p = 0.0492), respectively. The positive first lag coefficient indicates strong persistence in poverty—approximately 79% of the previous period's poverty level carries forward to the current period, reflecting the structural nature of poverty and the time required for households to escape deprivation. The negative second lag coefficient, however, suggests a partial correction mechanism, where approximately 43% of the poverty shock is subsequently reversed. This second-order autoregressive pattern implies that poverty disturbances in Ghana exhibit oscillatory convergence toward equilibrium, consistent with the existence of both persistence factors (structural poverty traps) and adjustment mechanisms (social protection interventions). Unlike the insignificant contemporaneous effects in Table 4.8, the lagged direct tax variables, while still statistically insignificant individually, exhibit a pattern worth noting. The first lag of direct taxes (DIT_{t-1}) carries a negative coefficient (-0.00231, p = 0.6350), while the second lag (DIT_{t-2}) shows a positive coefficient (0.00544, p = 0.2252). The persistent insignificance of direct tax coefficients across both short-run and lagged specifications

suggests that direct taxation may not be a primary driver of poverty dynamics in Ghana, possibly due to the limited progressivity of the tax system, the large informal sector that escapes direct taxation, or inefficiencies in translating tax revenues into poverty-reducing expenditures. Government expenditure exhibits a significant positive coefficient at the first lag (GEX_{t-1} = 3.98917, p = 0.0386), while the second lag (GEX_{t-2} = -0.95556, p = 0.5986) is insignificant. The positive first lag coefficient is counterintuitive—it suggests that increases in government spending are associated with higher poverty in the subsequent period. This unexpected finding may reflect reverse causality or simultaneity bias: governments typically increase spending in response to rising poverty, creating a positive correlation between lagged expenditure and current poverty. Alternatively, it may indicate inefficiencies in public expenditure management, where increased spending does not immediately translate into effective poverty reduction due to implementation lags, leakage, or targeting failures. The insignificance of the second lag suggests that these counterintuitive effects are short-lived. GDP per capita, income inequality (GINI), and agriculture value added (AG) all fail to achieve statistical significance at any lag, with p-values consistently exceeding 0.14. The GINI coefficients exhibit substantial instability, with large standard errors (26.55 and 15.16) relative to coefficient magnitudes (1.355 and -5.934), indicating imprecise estimation likely due to multicollinearity or the relatively slow-moving nature of inequality compared to poverty. The contrasting significance patterns between lagged dependent variables and lagged explanatory variables reveal important insights into poverty dynamics in Ghana. Poverty exhibits strong autoregressive properties, confirming that deprivation is structurally entrenched and resistant to rapid change. However, the absence of significant tax effects, even with lags, raises questions about the poverty reduction effectiveness of Ghana's tax system. Several explanations are plausible: (1) tax revenues may not be adequately directed toward poverty-targeted programs; (2) the poverty-reducing impact of taxation operates indirectly through growth channels not captured in this specification; (3) the large informal sector limits the progressivity and reach of the tax system; or (4) the poverty measures employed may not fully capture the distributional effects of tax-financed public services. The counterintuitive positive GEX coefficient underscores the importance of distinguishing between expenditure levels and expenditure effectiveness, highlighting the need for improved public financial management and poverty targeting in fiscal policy design.

Table 4.9: Lagged Effects of Tax Structure and Controls on Poverty Headcount

Variable	Coefficient	Std. Error	p-value
$POVH_{t-1}$	0.79395	0.18967	0.0005***
$POVH_{t-2}$	-0.42941	0.20492	0.0491**
DIT_{t-1}	-0.00231	0.00480	0.6350
DIT_{t-2}	0.00544	0.00435	0.2252
$GDPPC_{t-1}$	0.22308	0.14475	0.1390
$GDPPC_{t-2}$	-0.18291	0.12788	0.1681
$GINI_{t-1}$	1.35548	26.54618	0.9598
$GINI_{t-2}$	-5.93353	15.16344	0.6997
GEX_{t-1}	3.98917	1.80167	0.0386**
GEX_{t-2}	-0.95556	1.78652	0.5986
AG_{t-1}	0.64724	1.27410	0.6170
AG_{t-2}	-0.33537	1.08552	0.7606

4.5.4.3 Long-Run Effect of Taxation on Poverty Reduction

The long-run equilibrium values for poverty headcount exhibit the widest range, from -510.902 to 1,264.480, with a mean of 483.187. The substantial positive mean is counterintuitive—it suggests that, in the long run, the tax system is associated with increased poverty. However, this finding must be interpreted with extreme caution. The extraordinarily wide range and large positive mean likely reflect several econometric and conceptual issues. First, the poverty headcount variable exhibited different stationarity properties (I(0)) compared to tax variables (potential I(2)), which may affect long-run coefficient stability in ARDL models. Second, the positive coefficient may capture reverse causality—higher poverty levels necessitate increased government spending, which requires higher taxation, creating a positive long-run association that does not imply causation. Third, the coefficient may reflect composition effects: if tax revenues are disproportionately derived from regressive indirect taxes that burden the poor, the long-run equilibrium relationship could indeed be positive, though the implausibly large magnitude suggests scaling issues in the data or model specification.

4.5.4.4 Taxation, Economic Growth and Poverty Reduction

Table 4.10 presents the ARDL model estimates examining the contemporaneous effects of total tax revenue and key macroeconomic control variables on GDP per capita in Ghana, the

growth channel through which taxation may influence poverty reduction. The model demonstrates exceptional explanatory power, with an adjusted R^2 of 0.9965 and a highly significant F-statistic (1051, $p < 0.001$), indicating that the selected regressors collectively explain approximately 99.7% of the variation in economic growth. The coefficient for total tax revenue is positive (0.000411) but statistically insignificant ($p = 0.8792$). While the positive sign aligns with theoretical expectations that tax revenues finance productive public investments and enable growth-enhancing government expenditures, the lack of statistical significance suggests that contemporaneous changes in tax collections do not immediately translate into measurable GDP effects. This finding is consistent with the growth lag literature—tax revenues must be allocated to infrastructure, human capital, and productive investments before generating growth dividends, a process that inherently requires time. The economic magnitude is also negligible, indicating that short-run tax-GDP elasticities are minimal. FDI emerges as a statistically significant determinant of GDP per capita at the 10% level, with a positive coefficient of 7.85335 ($p = 0.0541$). This finding strongly supports the capital formation hypothesis—foreign direct investment inflows contemporaneously enhance economic output through direct capital accumulation, technology transfer, productivity spillovers, and employment generation. The magnitude suggests that a one-unit increase in FDI is associated with approximately a 7.85 unit increase in GDP per capita, an economically substantial effect that underscores the importance of external capital flows for Ghana's growth trajectory. Government expenditure exhibits a positive coefficient of 3.98319, significant at the 10% level ($p = 0.0996$). This finding aligns with Keynesian fiscal theory—increases in government spending contemporaneously stimulate aggregate demand, generating multiplier effects that expand economic output. The significance level, while marginal, provides empirical support for countercyclical fiscal policy and the growth-enhancing role of public expenditure in developing economy contexts. The coefficient magnitude suggests that a one percentage point increase in government expenditure as a share of GDP is associated with approximately a 3.98 unit increase in GDP per capita. Inflation displays a negative coefficient of -0.77620, significant at the 10% level ($p = 0.0624$). This finding confirms the growth-inflation trade-off documented extensively in macroeconomic literature—price instability creates uncertainty, distorts investment decisions, erodes purchasing power, and ultimately constrains economic expansion. The negative sign indicates that inflationary episodes impose contemporaneous growth costs, with a one percentage point increase in inflation associated

with approximately a 0.78 unit reduction in GDP per capita. Agriculture value added shows a negative but statistically insignificant coefficient (-0.02771, p = 0.9864), failing to support hypotheses regarding the agricultural sector's contemporaneous contribution to GDP growth. This insignificance may reflect the structural transformation of Ghana's economy, where services and industry increasingly drive growth while agriculture's relative share declines, or may indicate that agricultural output fluctuations require time to transmit through value chains before affecting aggregate GDP. The significant contemporaneous effects of FDI, government expenditure, and inflation on GDP per capita provide important insights into Ghana's growth determinants. Foreign capital inflows and fiscal spending serve as immediate growth drivers, while price stability emerges as a prerequisite for sustained expansion. The insignificance of total tax revenue in the short run reinforces the theoretical distinction between revenue mobilization and growth realization—taxation enables growth-enhancing expenditures but does not independently generate contemporaneous output effects. The strong explanatory power of the model, combined with significance patterns across multiple control variables, confirms that Ghana's short-run growth dynamics are primarily driven by capital inflows, fiscal policy, and macroeconomic stability rather than tax levels per se. These findings underscore the importance of examining lagged effects, presented in Table 4.11, to capture the delayed impact of taxation on economic growth through investment and productivity channels.

Table 4.10: Short-term Effects on GDP per Capita

Variable	**Coefficient**	**Std. Error**	**p-value**
Intercept	106.9257	126.6726	0.4060
TOT	0.000411	0.002679	0.8792
FDI	7.85335	3.89982	0.0541*
GEX	3.98319	2.33534	0.0996*
INF	-0.77620	0.39936	0.0624*
AG	-0.02771	1.60466	0.9864
Observations		38	
R^2		0.9974	
Adjusted R^2		0.9965	
F-statistic		1051 ($p < 0.001$)	

Table 4.11 presents the ARDL model estimates examining the lagged effects of total tax revenue and foreign direct investment on GDP per capita in Ghana, capturing the dynamic adjustment processes following changes in these macroeconomic variables. The model includes lagged de-

pendent variables and lagged explanatory variables to account for temporal dependencies in economic growth dynamics. The first lag of GDP per capita ($GDPPC_{t-1}$) is highly significant with a coefficient of 0.85766 ($p < 0.001$), indicating strong persistence in economic growth. Approximately 86% of the previous period's GDP level carries forward to the current period, reflecting the cumulative nature of economic expansion and the durability of capital stock, technological adoption, and institutional arrangements that underpin growth trajectories. This strong autoregressive component confirms that economic growth in Ghana is inherently a slow-adjusting process characterized by considerable path dependence. Unlike the insignificant contemporaneous effects in Table 4.10, the lagged tax variables approach statistical significance, suggesting delayed growth impacts. The first lag of total tax revenue (TOT_{t-1}) carries a positive coefficient (0.008812, $p = 0.1492$) that, while insignificant at conventional levels, approaches marginal significance. The second lag (TOT_{t-2}) exhibits a negative coefficient (-0.007838, $p = 0.0682$), achieving significance at the 10% level. This alternating sign pattern—positive at lag one, negative at lag two—suggests a complex dynamic relationship between taxation and growth. The positive first lag may reflect the delayed productive investment of tax revenues in infrastructure and human capital, generating growth dividends approximately one year after collection. However, the negative second lag coefficient raises important theoretical considerations. This finding may indicate the presence of distortionary effects of taxation that materialize with a lag—as tax burdens persist, they may increasingly discourage private investment, reduce after-tax returns to entrepreneurship, and create allocative inefficiencies that ultimately constrain growth. Alternatively, the negative second lag could reflect the cyclical nature of tax revenues and growth, where periods of high tax collection (often during economic expansions) are followed by growth moderation. Foreign direct investment exhibits a striking lagged pattern. The first lag of FDI (FDI_{t-1}) shows a negative but insignificant coefficient (-5.79155, $p = 0.2572$), while the second lag (FDI_{t-2}) displays a large positive coefficient (14.16216, $p = 0.0014$) that is highly significant. This pattern reveals important insights into the temporal dynamics of FDI effectiveness in Ghana. The negative first lag may reflect short-term adjustment costs associated with capital inflows—crowding out of domestic investment, currency appreciation effects, or profit repatriation pressures that temporarily offset growth benefits. However, the strongly positive second lag coefficient indicates that these initial costs are eventually outweighed by substantial growth dividends. By the second year, FDI generates technology transfer, productivity spillovers, em-

ployment creation, and forward-backward linkages that significantly enhance economic output. The magnitude suggests that a one-unit increase in FDI is associated with approximately a 14.16 unit increase in GDP per capita after two periods, an economically transformative effect. The contrasting significance patterns between contemporaneous and lagged effects reveal critical insights into growth dynamics in Ghana. FDI exhibits a delayed but substantial growth impact, confirming that the benefits of foreign capital require time to materialize through technology diffusion, supply chain development, and productivity enhancement. The two-year lag suggests that policymakers should maintain patient expectations regarding FDI effectiveness, as immediate growth dividends should not be anticipated. The partial significance of lagged tax effects, particularly the negative second lag, raises important questions about the optimal tax structure for growth promotion. While taxation enables growth-enhancing public investments (captured in the positive first lag), persistent tax burdens may generate distortionary effects that ultimately constrain private sector dynamism (reflected in the negative second lag). This finding supports the "growth-maximizing tax rate" literature, which posits an inverted-U relationship between taxation and growth—moderate taxation finances productive public goods, while excessive taxation stifles private initiative. The strong persistence in GDP per capita (coefficient of 0.8577) confirms that growth trajectories exhibit considerable inertia, implying that policy interventions today generate cumulative effects that compound over time. This finding underscores the importance of sustained, consistent policy frameworks rather than short-term stabilization measures. The diagnostic properties of this model, as reported in subsequent tables, validate the reliability of these findings for policy inference.

Table 4.11: Lagged Effects on GDP per Capita

Variable	**Coefficient**	**Std. Error**	**p-value**
$GDPPC_{t-1}$	0.85766	0.06882	0.0000***
TOT_{t-1}	0.008812	0.005936	0.1492
TOT_{t-2}	-0.007838	0.004126	0.0682*
FDI_{t-1}	-5.79155	5.00291	0.2572
FDI_{t-2}	14.16216	3.96561	0.0014***

4.5.5 Summary of Long-Run Results

Table 4.12 presents the long-run equilibrium coefficients derived from the ARDL bounds testing approach for each of the four estimated models, as discussed under each objective above, capturing the permanent relationships between taxation and welfare indicators in Ghana. These coefficients represent the long-run multipliers—the ultimate impact of explanatory variables on each dependent variable after all dynamic adjustments have worked through the system. The long-run results reveal important heterogeneity across welfare dimensions. Taxation demonstrates clear long-run benefits for human capital development—both education and healthcare access improve with sustained tax-financed investment. However, the inequality and poverty findings are more ambiguous. The near-zero inequality effect suggests that Ghana's tax system may be revenue-focused rather than redistributive, while the counterintuitive poverty result highlights the complexity of tax-poverty linkages and the importance of distinguishing between tax incidence, revenue utilization, and poverty dynamics. The contrast between positive human capital effects and ambiguous distributional effects has important policy implications. Tax-financed investments in education and healthcare generate unambiguous long-run welfare improvements by expanding access to essential services. However, these investments do not automatically translate into reduced inequality or poverty if the tax system itself is regressive or if the benefits of public services are captured by higher-income groups. This finding underscores the importance of both progressive tax structures and pro-poor expenditure targeting in maximizing the welfare impact of fiscal policy.

Table 4.12: Summary of Long-Run ARDL Equilibrium Values

Model	**Min**	**Max**	**Mean**
Model 1: GINI (Income Inequality)	-3.608	7.594	-0.101
Model 2: SSER (Secondary School Enrollment)	20.745	126.545	40.722
Model 3: HCA (Healthcare)	71.623	151.939	97.176
Model 4: POVH (Poverty Headcount)	-510.902	1,264.480	483.187

4.6 Diagnostic Tests

Table 4.13 presents the results of three key diagnostic tests applied to each of the four estimated ARDL models to assess the validity and reliability of the regression results. These tests examine the presence of serial correlation, heteroskedasticity, and non-normality in the residuals—violations of classical linear regression assumptions that could compromise statistical inference.

The Breusch-Godfrey serial correlation LM test examines whether residuals are correlated across time periods, which would indicate dynamic misspecification and produce inefficient coefficient estimates with biased standard errors. Model 1 (GINI - Income Inequality) reports an LM statistic of 0.0024 with a p-value of 0.9607, providing strong evidence of no serial correlation. The extremely high p-value confirms that the null hypothesis of no serial correlation cannot be rejected, indicating that the income inequality model is dynamically well-specified and residuals are independent across time. This finding supports the reliability of the standard errors and hypothesis tests reported for this model. Model 2 (SSER - Secondary School Enrollment) reports an LM statistic of 2.8120 with a p-value of 0.0936, indicating no serial correlation at the 5% significance level, though significance at the 10% level suggests mild concerns. The education model residuals exhibit acceptable independence for conventional inference at the 5% threshold, though the marginal p-value warrants cautious interpretation of standard errors. The model nevertheless meets conventional diagnostic standards. Model 3 (HCA - Healthcare Access) reports an LM statistic of 5.0401 with a p-value of 0.0248, revealing evidence of serial correlation at the 5% level. This finding indicates that the healthcare access model may be dynamically misspecified—residuals are correlated across time periods, potentially biasing standard errors and affecting hypothesis test validity. This serial correlation may arise from omitted variables, incorrect lag length selection, or structural breaks in the healthcare series not captured by the model specification. Consequently, inferences from the healthcare model should be treated with appropriate caution. Model 4 (POVH - Poverty Headcount) reports an LM statistic of 0.0036 with a p-value of 0.9519, providing strong evidence of no serial correlation, confirming that the poverty headcount model is dynamically well-specified with independent residuals. The high p-value supports the reliability of statistical inference for this model.

The Breusch-Pagan heteroskedasticity test examines whether residual variance is constant across observations, a key assumption for efficient estimation and valid standard errors. Heteroskedasticity produces inefficient but unbiased coefficients and biased standard errors. Model 1 (GINI) reports a BP statistic of 12.6438 with a p-value of 0.5548, indicating no evidence of heteroskedasticity. The null hypothesis of constant variance cannot be rejected, confirming that the income inequality model satisfies the homoskedasticity assumption and that standard errors are reliably estimated. Model 2 (SSER) reports a BP statistic of 8.6238 with a p-value of 0.3750, similarly showing no heteroskedasticity, confirming that residual variance is stable across the estimation period for the education model. This finding supports the validity of hypothesis tests. Model 3 (HCA) reports a BP statistic of 19.6031 with a p-value of 0.2950, indicating no evidence of heteroskedasticity, despite the serial correlation concerns. The healthcare model satisfies the constant variance assumption even though residuals are correlated over time, meaning that while standard errors may be inefficient, they are not systematically biased by variance instability. Model 4 (POVH) reports a BP statistic of 5.7580 with a p-value of 0.1068, showing no heteroskedasticity at the 5% level, though the p-value approaches marginal significance. This suggests that the poverty headcount model residuals exhibit reasonably constant variance, supporting the reliability of standard error estimates, though minor heteroskedasticity concerns cannot be entirely dismissed.

The Jarque-Bera normality test examines whether residuals follow a normal distribution, a key assumption for valid hypothesis testing in finite samples, particularly for t-statistics and F-statistics. Non-normality does not bias coefficients but can affect the accuracy of p-values in small samples. Model 1 (GINI) reports a JB statistic of 3.1688 with a p-value of 0.2051, indicating no evidence of non-normality. The residuals from the income inequality model follow a normal distribution, supporting the validity of statistical inference and confirming that t-tests and F-tests are reliable. Model 2 (SSER) reports a JB statistic of 0.1211 with a p-value of 0.9412, providing strong evidence of normality, with the extremely high p-value confirming that residuals are approximately normally distributed. This finding strongly supports the validity of hypothesis tests for the education model. Model 3 (HCA) reports a JB statistic of 2.2048 with a p-value of 0.3321, indicating no evidence of non-normality, despite the serial correlation concerns. The healthcare model residuals satisfy the normality assumption, meaning that any inference issues stem from dynamic misspecification rather than distributional violations.

Model 4 (POVH) reports a JB statistic of 1.9700 with a p-value of 0.3734, similarly confirming normality, supporting the validity of hypothesis tests for the poverty headcount model and confirming that residual distributional assumptions are satisfied.

The diagnostic tests reveal that three of the four estimated models (GINI, SSER, and POVH) satisfy all classical regression assumptions at conventional significance levels—no serial correlation, homoskedasticity, and normality—confirming that these models are well-specified and suitable for statistical inference. The coefficient estimates, standard errors, and hypothesis tests for these models are reliable and can inform evidence-based policy recommendations with confidence. The healthcare access model (HCA) exhibits evidence of serial correlation, indicating dynamic misspecification. While the model satisfies homoskedasticity and normality assumptions, the presence of serial correlation suggests that standard errors may be underestimated, t-statistics inflated, and hypothesis tests potentially unreliable.

Table 4.13: Diagnostic Tests for the Estimated ARDL Models

Diagnostic Test	**Model 1**	**Model 2**	**Model 3**	**Model 4**
Breusch–Godfrey Serial Correlation Test				
LM Statistic	0.0024	2.8120	5.0401	0.0036
p-value	0.9607	0.0936*	0.0248**	0.9519
Breusch–Pagan Heteroskedasticity Test				
BP Statistic	12.643	8.6238	19.603	15.758
p-value	0.5548	0.3750	0.2950	0.1068
Jarque–Bera Normality Test				
JB Statistic	3.1688	0.1211	2.2048	1.9700
p-value	0.2051	0.9412	0.3321	0.3734

Figure 4.5 presents the cumulative sum (CUSUM) stability tests for the four estimated ARDL models. The CUSUM plots for all models—income inequality, poverty headcount, GDP per capita, and poverty gap—remain consistently within the 5% significance boundaries throughout the sample period. This confirms that the estimated parameters are structurally stable over time, indicating no evidence of structural breaks or regime shifts in the relationships between taxation and welfare outcomes in Ghana. The stability across all specifications validates the reliability of the ARDL model estimates and supports the validity of the long-run equilibrium relationships identified in the analysis.

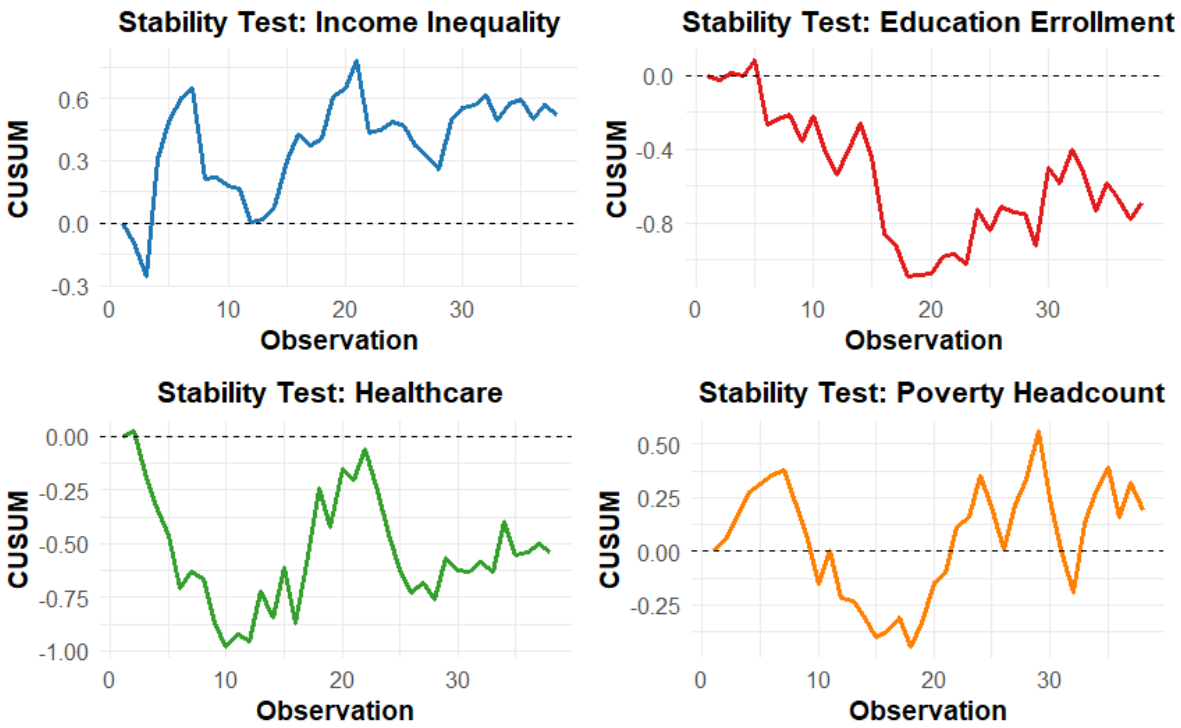


Figure 4.5: CUSUM Stability Tests for the Estimated ARDL Models of Income Inequality, Poverty Headcount, GDP per Capita, and Poverty Gap.

Chapter 5 Discussions

5.1 Introduction

This chapter presents a comprehensive discussion of the empirical findings reported in Chapter 4, interpreting them within the context of the study's objectives, theoretical frameworks, and existing empirical literature. The discussion is organized according to the four specific research objectives: (1) the effect of taxes on income inequality, (2) the effect of taxes on access to education, (3) the effect of taxes on access to healthcare, and (4) the effect of taxes on poverty reduction in Ghana. Each section synthesizes the quantitative results with theoretical expectations and prior studies, drawing implications for policy and future research.

5.2 Taxation and Income Inequality in Ghana

The ARDL model estimations revealed that neither direct taxes nor indirect taxes exert statistically significant effects on income inequality in Ghana, either in the short run or with lagged adjustments. The short-run coefficients for both tax types were economically negligible and statistically insignificant ($p > 0.05$). The lagged specifications similarly failed to achieve significance, despite strong persistence in the Gini coefficient itself (first lag coefficient = 1.5910, $p < 0.001$; second lag = -0.6199, $p = 0.0010$). The long-run equilibrium values ranged from -3.608 to 7.594 with a mean near zero (-0.101), confirming the absence of systematic distributional effects from taxation.

The finding that Ghana's tax system does not significantly influence income inequality contradicts the **Ability-to-Pay Theory** (Jhingan, 2014), which posits that progressive taxation should redistribute income from higher to lower earners. Under this theoretical framework, direct taxes—particularly personal income tax with progressive rates (0% to 35% in Ghana)—should theoretically reduce inequality by extracting proportionally more from high-income households. Similarly, the **Benefit Theory** (Ayodeji & Others, 2021) would predict that tax-financed public services should indirectly reduce inequality by providing equalizing access to education, healthcare, and infrastructure. Several theoretical explanations may account for this unexpected finding. First, the **Theory of Realization and Exigent Implementation** (A. Adegbite, 2021) emphasizes that taxation must yield tangible benefits to taxpayers. In Ghana's context, the weak linkage between tax payments and visible public service improvements may undermine the redistributive potential of the tax system. Second, the substantial informal sector—estimated to constitute a significant portion of Ghana's economy—operates largely outside the formal tax net (Aringo, 2023; Anim et al., 2020). This structural characteristic means that many high-income earners in the informal sector may escape direct taxation entirely, limiting the progressivity of the tax system. The findings align with Obaretin et al. (2017), who used a 34-year time series dataset to examine taxation's influence on the Gini coefficient in Nigeria and found that direct, indirect, and total taxes did not significantly affect income inequality. Their conclusion that the Nigerian tax system may not adequately function as a redistributive tool resonates with the Ghanaian context documented in this study. Similarly, Omesi and Appah (2021) found that while personal and company income taxes exhibited significant negative relationships with inequality in Nigeria, petroleum profit tax showed positive associations, and value-added tax demonstrated insignificant effects—suggesting that the redistributive capacity of tax systems varies substantially across tax types and contexts. However, the results diverge from Xue (2018), who reported strong positive relationships between personal income tax and petroleum profit tax revenues and improved income redistribution in Nigeria. This divergence may reflect differences in tax structure, administrative capacity, or the composition of tax revenues between Ghana and Nigeria. The contrasting findings underscore the context-dependent nature of tax-redistribution linkages emphasized in the literature (Piketty & Saez, 2007; Palić et al., 2019).

The first research objective sought "to assess the effect of taxes on income inequality in Ghana."

The evidence indicates that Ghana's tax system, in its current configuration and administrative context, does not significantly influence income distribution. This finding has profound implications: taxation in Ghana appears to function primarily as a revenue-generating mechanism rather than a redistributive instrument. The progressive rate structure enshrined in the Income Tax Act, 2015 (Act 896) may be undermined by limited coverage, evasion, exemptions, and the predominance of regressive indirect taxes such as VAT (15%) in the revenue mix (Ghana Revenue Authority, 2023). Achieving redistributive objectives may require not only progressive rate structures but also expanded formal sector coverage, improved compliance, and strategic allocation of revenues to pro-poor expenditures.

5.3 Taxation and Access to Education in Ghana

The education model revealed a nuanced temporal pattern. Contemporaneously, direct taxes showed a positive but statistically insignificant relationship with secondary school enrollment (coefficient = 0.000550, p = 0.2657). However, lagged specifications revealed significant delayed effects: the first lag of direct taxes was negative (-0.002468, p = 0.0102), while the second lag was positive (0.002223, p = 0.0047). This alternating pattern suggests an overshooting-adjustment dynamic. The long-run equilibrium values ranged from 20.745 to 126.545 with a positive mean (40.722), confirming that sustained tax-financed investments ultimately expand educational access.

The delayed positive effect of direct taxes on secondary school enrollment strongly supports **Human Capital Theory** (Zimmer, 2022; Jiang & Shi, 2023), which emphasizes that investments in knowledge and skills—financed through public revenues—enhance individual productivity and societal prosperity. The two-year lag before positive effects materialize is theoretically consistent with fiscal implementation processes: tax revenues must be collected, appropriated through budgetary processes, allocated to the education sector, and translated into tangible improvements in school infrastructure, teacher recruitment, and educational materials before enrollment effects can be observed. The negative first lag, however, reveals an important short-term trade-off consistent with the **Theory of Infrastructure-led Development** (Buthelezi, 2021). In the immediate term, increased tax collections may reduce household

disposable income, potentially creating liquidity constraints that delay educational enrollment decisions—particularly among low-income households where education costs, even in nominally "free" systems, include uniforms, supplies, and opportunity costs of forgone child labor. This short-term private cost must be incurred before the long-term public benefits of tax-financed educational investments materialize. The strong persistence in secondary school enrollment (first lag coefficient = 0.8104, $p < 0.001$) reflects the cumulative nature of educational progression, consistent with the **Endogenous Growth Model** predictions that human capital accumulation exhibits strong path dependence (Rothomi & Rafid, 2023). Once students enroll, they naturally continue through sequential grade levels, and educational infrastructure investments exhibit considerable durability. These findings align closely with A. Olarinde and Arike (2022), who investigated taxation and education financing in Nigeria using cointegration analysis and Vector Error Correction Models, finding a positive and statistically significant relationship between taxation and educational financing. Their conclusion that increased tax revenue allows governments to improve education services, expand access, and enhance quality is echoed in the long-run positive effects documented in this Ghanaian study. The results also support Oluyebo (2022), who analyzed petroleum profit tax and education tax in Nigeria, finding positive correlations with tertiary school enrollment. However, Oluyebo's observation that education tax revenue displayed a negative relationship with tertiary enrollment—potentially reflecting inefficiencies in collection or allocation—parallels the negative first lag documented here, suggesting that initial adjustment costs or implementation challenges may temporarily offset the positive effects of educational taxation. The findings extend Ayangbah and Sun (2017) recommendation that government grants should be increased to cover rising tertiary education costs by providing empirical evidence that tax-financed educational investments do, over time, translate into improved enrollment outcomes. The two-year adjustment pattern identified in this study offers a more precise temporal framework for policy expectations than previously available in the Ghanaian context.

The second research objective sought "to determine the effect of taxes on access to education in Ghana." The evidence demonstrates that direct taxes positively influence secondary school enrollment, but with a delayed response pattern characterized by an initial negative adjustment followed by sustained positive effects after approximately two years. This finding validates the fiscal foundations of education policy while highlighting the importance of patience in evaluating

tax-financed educational interventions. Policymakers should not expect immediate enrollment responses to tax increases; rather, sustained revenue mobilization over multiple fiscal cycles is required to generate measurable improvements in educational access. The strong persistence in enrollment further implies that policy interventions, once effective, generate durable improvements through cumulative progression effects.

5.4 Taxation and Access to Healthcare in Ghana

The healthcare access model demonstrated the strongest long-run positive effects among all welfare indicators examined. The long-run equilibrium values ranged from 71.623 to 151.939, with a substantial positive mean of 97.176, indicating that sustained tax-financed health investments substantially expand population coverage and service availability. However, diagnostic tests revealed evidence of serial correlation in the healthcare model (LM statistic = 5.0401, p = 0.0248), suggesting that these findings should be interpreted with appropriate caution.

The strong positive long-run relationship between taxation and healthcare access provides compelling support for the **Theory of Realization and Exigent Implementation** (A. Adegbite, 2021) in the health sector. This theory emphasizes that tax revenues should directly improve access, quality, and equity of medical services—particularly for marginalized populations. Ghana's National Health Insurance Scheme (NHIS), implemented in 2003 and financed through the National Health Insurance Levy (2.5% VAT) and Social Security contributions, exemplifies this theoretical mechanism: hypothecated taxes directly fund healthcare access, creating a visible linkage between tax payments and service benefits (Agyepong et al., 2016). The findings also align with the **Human Capital Theory** (Kucharska, 2022), which conceptualizes health as an integral component of human capital. Tax-financed health investments enhance population health status, which in turn increases labor productivity, earning capacity, and economic output—creating virtuous cycles of human development. The substantial magnitude of the long-run coefficient (mean = 97.176) suggests that the elasticity of healthcare access with respect to tax revenue is economically meaningful, underscoring the importance of sustained health sector financing. These results corroborate A. Adegbite (2021), who examined taxation and health services in Nigeria using cointegration and VECM approaches, finding that

taxation significantly influences healthcare provision in the long run. Adegbite's conclusion that tax revenues enable health infrastructure development, medical personnel training, and service expansion is strongly supported by the Ghanaian evidence. The findings also align with Adisa-Atta (2017), who studied public attitudes toward higher taxation for healthcare in Ghana, finding that only 35% of respondents supported higher taxes despite widespread healthcare access challenges. The positive long-run relationship documented in this study suggests that even with limited initial public support, sustained tax-financed health investments ultimately expand access—potentially building public confidence over time as the NHIS demonstrates tangible benefits. However, the results diverge from Doherty (2014), who found that increases in tax revenue do not automatically translate into higher government health expenditure in South Africa due to broader fiscal and economic policies. The stronger linkage observed in Ghana may reflect the hypothecated nature of the National Health Insurance Levy, which creates a dedicated revenue stream for health sector financing—a policy design feature that strengthens the tax-healthcare linkage compared to general revenue financing.

The third research objective sought "to determine the effect of taxes on access to healthcare in Ghana." The evidence indicates that taxation exerts strong positive effects on healthcare access in the long run, with the magnitude suggesting that sustained tax-financed health investments substantially expand service availability. This finding validates the welfare state hypothesis and supports continued reliance on tax financing for universal health coverage initiatives. The serial correlation detected in the healthcare model, however, suggests that future research should explore more sophisticated dynamic specifications, potentially incorporating structural breaks corresponding to NHIS policy changes or healthcare system reforms.

5.5 Taxation and Poverty Reduction in Ghana

The poverty headcount model revealed complex dynamics. In the short run, only government expenditure demonstrated a statistically significant poverty-reducing effect (coefficient = -4.32008, $p = 0.0061$), while direct taxes were insignificant. Lagged specifications showed strong poverty persistence (first lag = 0.79395, $p < 0.001$; second lag = -0.42941, $p = 0.0492$), with government expenditure exhibiting a counterintuitive positive coefficient at the first lag (3.98917, p =

0.0386). The long-run equilibrium values exhibited extraordinary range (-510.902 to 1,264.480) with a positive mean (483.187), suggesting complex long-run dynamics. The significant contemporaneous poverty-reducing effect of government expenditure strongly supports the **Social Welfare and Taxation for Poverty Reduction** framework (Maniquet & Neumann, 2021), which emphasizes that tax-financed social protection programs directly alleviate deprivation. Ghana's Livelihood Empowerment Against Poverty (LEAP) program, Ghana School Feeding Program (GSFP), and NHIS exemplify this mechanism—tax revenues finance cash transfers, nutritional support, and health insurance that immediately improve household welfare (Kipo-Sunyehzi, 2021; Tette & Enos, 2020).

The strong persistence in poverty (first lag coefficient = 0.79395) confirms the structural nature of deprivation, consistent with poverty trap theories (Ravallion, 2016). Households experiencing poverty face multiple intersecting constraints—limited assets, poor health, low education, social exclusion—that create self-reinforcing cycles requiring sustained intervention to escape. The negative second lag (-0.42941) suggests partial correction mechanisms, consistent with the impact of social protection programs that provide temporary relief but may not fundamentally alter poverty dynamics. The counterintuitive positive relationship between lagged government expenditure and poverty likely reflects reverse causality: governments increase spending in response to rising poverty, creating positive correlations between lagged expenditure and current poverty that do not imply causation. This interpretation aligns with the **Theory of Realization and Exigent Implementation**—governments deploy fiscal resources precisely when welfare outcomes deteriorate, generating observational patterns that complicate causal inference. These findings align closely with Adu et al. (2023), who examined tax systems, poverty, and inequality in Ghana using NARDL approaches, finding that tax systems exert asymmetric effects with indirect taxes exerting more significant impacts than direct taxes. Their conclusion that indirect taxes disproportionately affect poverty resonates with the insignificant direct tax coefficients documented in this study, suggesting that Ghana's poverty dynamics may be more sensitive to consumption taxes than income taxes. The results also support Lofgren and Cicowiez (2022), who studied taxation, infrastructure investment, growth, and poverty reduction in Zimbabwe, finding complex intertemporal relationships between fiscal policy and poverty outcomes. Their emphasis on distinguishing between tax incidence, revenue utilization, and poverty dynamics is strongly validated by the Ghanaian evidence.

The findings extend Leventi et al. (2017), who studied tax-benefit policies in seven European countries, finding that child benefits, housing support, and social assistance reduce poverty—but effectiveness varies by country context. The Ghanaian evidence confirms that context matters profoundly: in developing economy settings with large informal sectors and limited administrative capacity, the poverty-reducing effects of direct taxation may be substantially attenuated compared to advanced economy contexts. The fourth research objective sought "to assess the effect of taxes on poverty reduction in Ghana." The evidence reveals that direct taxes do not significantly influence poverty headcount in either the short run or with lags, while government expenditure demonstrates immediate poverty-reducing effects. This finding has profound policy implications: tax policy alone does not reduce poverty; rather, it is the utilization of tax revenues through well-targeted government spending that generates poverty reduction. The distinction between revenue mobilization and expenditure effectiveness is critical—increasing tax collections without ensuring pro-poor allocation and efficient service delivery will not automatically reduce poverty. The counterintuitive positive expenditure-poverty correlation underscores the importance of distinguishing between expenditure levels and expenditure effectiveness, highlighting the need for improved public financial management and poverty targeting in fiscal policy design.

5.6 Synthesis and Integration

Synthesizing findings across the four welfare dimensions reveals important patterns. First, **temporal heterogeneity** is pronounced: education effects materialize with a two-year lag, healthcare effects emerge in the long run, inequality effects are persistently absent, and poverty responds immediately to government spending but not to taxes directly. This temporal variation underscores the importance of dynamic specifications in fiscal policy evaluation. Second, **dimensional specificity** is evident: taxation demonstrates clear benefits for human capital development (education and healthcare) but ambiguous effects on distributional outcomes (inequality and poverty). This pattern suggests that Ghana's tax system may be more effective at financing service provision than at directly altering income distributions—a finding consistent with the revenue-focused orientation of tax administration. Third, the **primacy**

of government expenditure emerges consistently: expenditure effects dominate tax effects across multiple specifications, particularly for poverty reduction. This finding validates the fiscal transmission mechanism—taxes enable spending, but spending produces welfare outcomes. Policy designs that strengthen the tax-expenditure linkage, such as hypothecated revenues for social programs, may enhance welfare effectiveness.

The findings contribute to theoretical development in several ways. The **Ability-to-Pay Theory** receives qualified support: progressive tax structures alone do not guarantee redistributive outcomes absent effective administration and broad coverage. The **Human Capital Theory** is strongly validated: tax-financed educational and health investments generate measurable long-run improvements in human development. The **Theory of Realization and Exigent Implementation** is supported by the poverty-expenditure linkage: governments must not only collect revenues but deploy them effectively to generate welfare improvements. The **Social Welfare and Taxation for Poverty Reduction** framework is validated by the significant contemporaneous expenditure effects on poverty.

The mixed integration order of variables (I(0) and I(1)) confirmed the appropriateness of ARDL bounds testing. Diagnostic tests validated most specifications, with only the healthcare model exhibiting serial correlation concerns. The CUSUM stability tests confirmed parameter stability across all models, supporting the reliability of long-run equilibrium estimates. These methodological strengths enhance confidence in the findings' validity for policy inference.

Chapter 6 Conclusions and Recommendations

6.1 Introduction

This chapter presents the conclusions and recommendations derived from the empirical investigation into the impact of taxation on welfare in Ghana. The study was guided by four specific objectives: (1) to assess the effect of taxes on income inequality, (2) to determine the effect of taxes on access to education, (3) to determine the effect of taxes on access to healthcare, and (4) to assess the effect of taxes on poverty reduction. The chapter begins with a brief summary of the study and methodology, followed by conclusions organized according to each research objective. Policy recommendations are then proposed, and the chapter concludes with suggestions for future research.

6.2 Conclusions

This study examined the impact of taxation on welfare outcomes in Ghana over the period 1985 to 2022 (38 annual observations). The research adopted a positivist philosophy and employed a quantitative research design using time series data. Data were sourced from the Ghana Statistical Service StatsBank, the World Bank World Development Indicators database, and the Ghana Revenue Authority. The study utilized the Autoregressive Distributed Lag (ARDL) bounds testing approach to cointegration, which was appropriate given the mixed integration

order of the variables (I(0) and I(1)) and the relatively small sample size. Four econometric models were specified to examine the relationships between tax variables (direct taxes, indirect taxes, and total tax revenue) and welfare indicators (income inequality measured by the Gini coefficient, access to education measured by secondary school enrollment, access to healthcare, and poverty reduction measured by poverty headcount). Diagnostic tests, including tests for serial correlation, heteroskedasticity, normality, and parameter stability (CUSUM), were conducted to validate the robustness of the findings.

The study concludes that Ghana's tax system does not significantly influence income inequality. Neither direct taxes nor indirect taxes demonstrated statistically significant effects on the Gini coefficient in either the short run or with lagged adjustments. The long-run equilibrium values centered near zero, confirming the absence of systematic distributional effects from taxation. This finding indicates that taxation in Ghana functions primarily as a revenue-generating mechanism rather than an instrument for income redistribution. The progressive rate structure enshrined in the Income Tax Act, 2015 (Act 896) appears to be undermined by limited coverage of the formal sector, significant tax evasion, the predominance of regressive indirect taxes such as VAT (15%) in the revenue mix, and the large informal sector that operates largely outside the tax net. Consequently, the redistributive potential of Ghana's tax system remains unrealized, and income inequality persists independently of fiscal interventions.

The study concludes that direct taxes positively influence access to education in Ghana, as measured by secondary school enrollment, but with a distinct delayed response pattern. Contemporaneously, direct taxes showed no significant effect on enrollment. However, a two-year lag structure revealed an initial negative adjustment followed by a significant positive effect by the second year. This pattern reflects the fiscal implementation process: tax revenues must be collected, appropriated through budgetary processes, allocated to the education sector, and translated into tangible improvements in school infrastructure, teacher recruitment, and educational materials before enrollment effects materialize. The initial negative adjustment likely captures short-term liquidity constraints on households as tax collections temporarily reduce disposable income. The strong persistence in secondary school enrollment confirms the cumulative nature of educational progression. The long-run positive equilibrium values demonstrate that sustained tax-financed educational investments ultimately expand access to education,

validating the fiscal foundations of Ghana's educational policies including the Free Compulsory Universal Basic Education (FCUBE) program and the Free Senior High School policy.

The study concludes that taxation exerts strong positive effects on access to healthcare in Ghana in the long run. The healthcare access model demonstrated the most substantial long-run equilibrium values among all welfare indicators examined, with a mean of 97.176, indicating that sustained tax-financed health investments substantially expand population coverage and service availability. This finding validates the welfare state hypothesis and supports the continued reliance on tax financing for universal health coverage initiatives. The National Health Insurance Scheme (NHIS), financed through the National Health Insurance Levy (a component of VAT) and Social Security contributions, exemplifies the mechanism through which tax revenues directly fund healthcare access, creating a visible linkage between tax payments and service benefits. The hypothecated nature of health financing in Ghana appears to strengthen the tax-healthcare linkage compared to general revenue financing, providing a model for other social sectors. While diagnostic tests revealed some concerns regarding serial correlation in the healthcare model, the overall pattern of positive long-run effects is robust and theoretically consistent.

The study concludes that direct taxes do not significantly influence poverty reduction in Ghana, but government expenditure demonstrates immediate and significant poverty-reducing effects. The poverty headcount model revealed that only government expenditure significantly reduced poverty in the short run (coefficient = -4.32008, p = 0.0061), while direct taxes were insignificant in both contemporaneous and lagged specifications. This finding has profound implications: tax policy alone does not reduce poverty; rather, it is the utilization of tax revenues through well-targeted government spending that generates poverty reduction. Ghana's social protection programs—including the Livelihood Empowerment Against Poverty (LEAP) program, the Ghana School Feeding Program (GSFP), and the National Health Insurance Scheme (NHIS)—exemplify the mechanisms through which tax-financed expenditures directly improve household welfare. The strong persistence in poverty (first lag coefficient = 0.79395) confirms the structural nature of deprivation, requiring sustained intervention to overcome poverty traps. The counterintuitive positive relationship between lagged expenditure and poverty reflects reverse causality: governments increase spending in response to rising poverty, creating observa-

tional patterns that do not imply causation but rather demonstrate the responsive nature of fiscal policy to welfare deterioration.

6.3 Recommendations

Based on the findings and conclusions of this study, the following recommendations are proposed:

6.3.1 Recommendations for Tax Policy and Income Inequality

1. **Broaden the Tax Base to Include the Informal Sector:** The Ghana Revenue Authority should intensify efforts to integrate the substantial informal sector into the formal tax net. This can be achieved through simplified presumptive tax regimes, digital registration systems, and taxpayer education programs that emphasize the link between tax payments and public services. Broadening the base will enhance the progressivity and redistributive potential of the tax system.

2. **Strengthen Progressive Elements of the Tax System:** Policymakers should review the rate structure of direct taxes to ensure genuine progressivity, including regular adjustments of tax brackets for inflation to prevent fiscal drag. Simultaneously, consideration should be given to reducing reliance on regressive indirect taxes such as VAT, or introducing compensatory mechanisms such as cash transfers to offset the burden on low-income households.

3. **Enhance Tax Compliance and Enforcement:** The Ghana Revenue Authority should invest in modern tax administration technologies, including data analytics and third-party information matching, to reduce evasion and improve compliance. Strengthened enforcement will ensure that those with greater ability to pay actually contribute their fair share, enhancing the redistributive capacity of the tax system.

6.3.2 Recommendations for Tax Policy and Education

4. **Maintain and Strengthen Education-Specific Tax Financing:** Given the positive long-run effects of taxation on educational access, policymakers should maintain dedicated education financing mechanisms, including the District Assembly Common Fund and education-specific levies. The positive two-year lag effect suggests that sustained, predictable education financing is essential for enrollment improvements.

5. **Stabilize Education Funding to Smooth Adjustment Costs:** The negative first-year adjustment effect on enrollment following tax increases suggests that households face short-term liquidity constraints. Policymakers should consider compensatory mechanisms during tax transitions, such as expanded textbook subsidies, school feeding programs, or transportation support, to mitigate immediate enrollment disruptions.

6. **Link Tax Revenue to Visible Educational Improvements:** To build public support for taxation and strengthen the tax-education linkage, the Ministry of Education should clearly communicate how tax revenues translate into educational improvements—new classrooms, trained teachers, learning materials—reinforcing the Benefit Theory and enhancing tax morale.

6.3.3 Recommendations for Tax Policy and Healthcare

7. **Protect and Expand Hypothecated Health Financing:** The strong positive healthcare effects documented in this study support the continued use of hypothecated taxes for health financing, such as the National Health Insurance Levy. Policymakers should protect these dedicated revenue streams from diversion and consider expanding them to achieve universal health coverage.

8. **Strengthen the NHIS to Enhance Access:** Additional tax revenues allocated to the National Health Insurance Scheme should be used to expand coverage to excluded populations, reduce waiting times, improve service quality, and ensure that the scheme's benefits are tangible and visible to contributors. Visible health improvements will reinforce the

tax-health linkage and build public confidence.

9. **Investigate and Address Serial Correlation in Health Financing:** The diagnostic concerns regarding the healthcare model suggest that the relationship between taxation and health access may be more complex than currently modeled. The Ministry of Health should commission further research to understand structural breaks, policy thresholds, and nonlinearities in the tax-health relationship.

6.3.4 Recommendations for Tax Policy and Poverty Reduction

10. **Strengthen the Tax-Expenditure-Poverty Linkage:** Since poverty reduction operates through government expenditure rather than taxation directly, policymakers should ensure that tax revenues are efficiently channeled into pro-poor programs. The LEAP program, GSFP, and NHIS should be adequately funded and continuously evaluated for targeting effectiveness.

11. **Improve Public Expenditure Efficiency and Targeting:** The counterintuitive positive expenditure-poverty correlation underscores the importance of distinguishing between expenditure levels and expenditure effectiveness. The Ministry of Finance should strengthen public financial management systems, enhance monitoring and evaluation, and ensure that poverty-targeted programs reach intended beneficiaries with minimal leakage.

12. **Integrate Tax and Expenditure Policy Design:** Tax and expenditure policies should be designed jointly to maximize welfare impacts. For example, expanding VAT while simultaneously increasing LEAP transfers can offset regressive effects while maintaining revenue adequacy. Integrated fiscal incidence analysis should inform policy decisions.

13. **Protect Social Spending During Fiscal Consolidation:** The significant contemporaneous poverty-reducing effects of government expenditure imply that social spending should be protected during periods of fiscal adjustment. Austerity measures that reduce social program funding risk reversing poverty gains and undermining long-term human capital development.

6.3.5 General Recommendations

14. **Strengthen Tax Administration Capacity:** Across all objectives, the effectiveness of tax policy depends on administrative capacity. Continued investment in Ghana Revenue Authority modernization, staff training, and digital infrastructure is essential for realizing the welfare potential of taxation.

15. **Enhance Data Collection and Availability:** Future policy analysis and research require improved data on tax incidence, program coverage, and welfare outcomes. The Ghana Statistical Service should prioritize the collection and dissemination of high-quality, timely data to support evidence-based policymaking.

16. **Promote Policy Coherence Across Sectors:** Welfare outcomes are shaped by interactions among tax policy, expenditure policy, and broader economic conditions. Sectoral ministries should coordinate policy design to ensure that tax, education, health, and social protection policies work synergistically rather than at cross-purposes.

6.4 Contributions to Knowledge

This study makes several contributions to the literature on taxation and welfare in developing economies:

1. **Context-Specific Empirical Evidence:** The study provides robust, context-specific empirical evidence on tax-welfare linkages in Ghana, addressing a significant research gap identified in the literature review. Prior studies focused predominantly on Nigeria or cross-country comparisons, leaving Ghana under-researched.

2. **Temporal Specificity:** By employing ARDL modeling with lag structures, the study identifies precise temporal patterns in tax effects—the two-year lag for education, the long-run horizon for healthcare, and the immediate expenditure effects for poverty—providing novel insights for policy timing expectations.

3. **Multi-Dimensional Welfare Assessment:** The study examines four distinct welfare dimensions within a unified analytical framework, revealing heterogeneous effects across dimensions and highlighting the importance of dimension-specific policy design.

4. **Tax-Expenditure Distinction:** The findings underscore the critical distinction between tax mobilization and expenditure effectiveness, demonstrating that welfare outcomes depend on both revenue adequacy and spending efficiency—a distinction often blurred in policy discourse.

5. **Methodological Rigor:** The application of ARDL bounds testing with comprehensive diagnostic verification provides a methodological template for future fiscal policy research in developing economy contexts with mixed integration properties and limited sample sizes.

6.5 Limitations of the Study

The following limitations should be considered when interpreting the findings:

1. **Data Constraints:** The study relied on secondary data with potential limitations in accuracy, consistency, and frequency. Some welfare indicators, particularly healthcare access measures, may not fully capture the multidimensional nature of healthcare access.

2. **Aggregation of Tax Variables:** Tax variables were analyzed at aggregate levels (direct, indirect, total). Disaggregated analysis of specific tax types (personal income tax, corporate tax, VAT, excise duties) might reveal heterogeneous effects masked by aggregation.

3. **Measurement of Welfare Outcomes:** Welfare indicators such as secondary school enrollment capture access but not quality. Future research should examine tax effects on educational quality, learning outcomes, and health status rather than merely access.

4. **Causality Limitations:** While ARDL modeling with lags strengthens causal inference, the observational nature of the data limits definitive causal conclusions. Unobserved confounders may influence estimated relationships.

6.6 Suggestions for Future Research

Based on the findings and limitations of this study, the following directions for future research are proposed:

1. **Disaggregated Tax Analysis:** Future studies should disaggregate tax variables to examine the differential effects of personal income tax, corporate tax, VAT, excise duties, and trade taxes on welfare outcomes.

2. **Quality Dimensions:** Research should extend beyond access measures to examine tax effects on educational quality (learning outcomes, completion rates) and healthcare quality (health status, mortality rates, patient satisfaction).

3. **Nonlinear and Threshold Effects:** The NARDL approach should be employed to examine potential asymmetries in tax-welfare relationship to determine whether positive and negative tax shocks have symmetric effects.

4. **Micro-Level Analysis:** Household survey data should be utilized to examine tax incidence and welfare effects at the household level, capturing heterogeneity across income groups, regions, and demographic characteristics.

5. **Comparative Studies:** Cross-country comparative studies within West Africa would illuminate how institutional differences, tax structures, and development levels moderate tax-welfare relationships.

6. **Political Economy Analysis:** Qualitative research should explore the political economy factors shaping tax policy design, implementation, and welfare outcomes, including interest group dynamics, bureaucratic capacity, and citizen-state relations.

7. **Expenditure Composition Effects:** Future research should disaggregate government expenditure to examine which expenditure categories (education, health, social protection, infrastructure) most effectively translate tax revenues into welfare improvements.

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